LM01 Alternative Investment Features, Methods, and Structures

1. Introduction....2
2. Alternative Investment Features....2
3. Alternative Investment Methods....4
4. Alternative Investment Structures....6
Summary....9

This document should be read in conjunction with the corresponding learning module in the 2024 Level I CFA® Program curriculum. Some of the graphs, charts, tables, examples, and figures are copyright 2023, CFA Institute. Reproduced and republished with permission from CFA Institute. All rights reserved.

Required disclaimer: CFA Institute does not endorse, promote, or warrant the accuracy or quality of the products or services offered by IFT. CFA Institute, CFA®, and Chartered Financial Analyst® are trademarks owned by CFA Institute.

Version 1.0

© IFT. All rights reserved

1. Introduction

This learning module covers:

- Features and categories of alternative investments
- Methods of investing in alternative investments; the advantages and disadvantages of each method
- Investment and compensation structures commonly used in alternative investments

Traditional investments refer to long-only positions in stocks, bonds, and cash. All other investments are classified as alternative investments.

Alternative investments can be divided into three main categories:

- Private capital
- Real assets
- Hedge funds

2. Alternative Investment Features

Why Investors Consider Alternative Investments

Since the mid-1990s assets under management in alternative investments have grown significantly. Investors consider alternative investments due to:

- The potential for portfolio diversification. Alternative investments have low correlation with traditional asset classes.
- The opportunities for enhanced returns. Adding alternative investments can increase the portfolio's risk-return profile.
- The potentially increased income through higher yields. During low-interest rate periods, alternative investments can provide significantly higher yields as compared to traditional investments.

Alternative Investments: Features and Categories

Some features of alternative investments are same as traditional investments, while some features are significantly different. Features that distinguish alternative investments are:

- Need for specialized knowledge: For example, within private equity, you have leveraged buyout and venture capital. There are managers who focus only on leveraged buyouts within private equity.
- Relatively low correlation with traditional investments. But correlation may increase during times of financial crisis.
- Illiquidity, long investment time horizons, and large capital outlays

Due to these features, alternative investments have the following characteristics:

- Different investment structures to solve the challenges of direct investment

© IFT. All rights reserved

- Incentive-based fee structure to address information asymmetry between managers and investors.
- Performance appraisal challenges

Categories of Alternative Investments

The main categories of alternative investments are:

Private capital

Private equity:

- Private equity funds invest in the equity of private companies or public companies that want to become private.
- They are further divided into
  - *Leveraged buyout funds*: Invest in established companies.
  - *Venture capital funds*: Invest in startups or early-stage companies.

Private debt:

- Includes debt provided to private entities.
- Forms of private debt include:
  - *Direct lending*: private loans with no intermediary.
  - *Mezzanine loans*: private subordinated debt.
  - *Venture debt*: private loans to startups or early-stage companies.
  - *Distressed debt*: private loans to distressed companies, e.g., companies facing bankruptcy.

Real Assets

Real estate:

- Investments in buildings or land either directly or indirectly.
- Securitization has broadened the definition of real estate investing and it now includes:
  - *Private commercial real estate equity*: e.g., ownership of an office building
  - *Private commercial real estate debt*: e.g., directly issued mortgages on commercial property
  - *Public real estate equity*: e.g., real estate investment trusts (REITS)
  - *Public real estate debt*: e.g., mortgage-backed securities (MBS).

Infrastructure:

- Investments in capital intensive, long-lived, real assets such as roads, dams, and schools, which are intended for public use and provide essential service.
- A common approach to infrastructure investing is a public–private partnership (PPP) in which both the government and investors have a stake.

© IFT. All rights reserved

Natural Resources:

Commodities:

- Investment in physical assets such as grains, metals, crude oil, etc.
- Commodity investments can be done by either owning physical assets, using derivative products, or investing in business engaged in the exploration and production of physical commodities.

Agricultural land (or farmland):

- Investments in land used for the cultivation of crops or livestock.
- Income can be generated from the growth, harvest and sale of crops or livestock; or by leasing the land back to farmers.

Timberland:

- Investments in natural forests or managed tree plantations.
- The return comes from the sale of trees, wood, and other timber products.

Digital Assets:

- An umbrella term that covers assets that can be created, stored, and transmitted electronically an have associated ownership or use rights.
- E.g., cryptocurrencies and tokens.

Hedge funds

- They are private investment vehicles that manage portfolios of securities and derivative positions using a variety of strategies.
- Some hedge funds aim for absolute returns independent of market performance.

3. Alternative Investment Methods

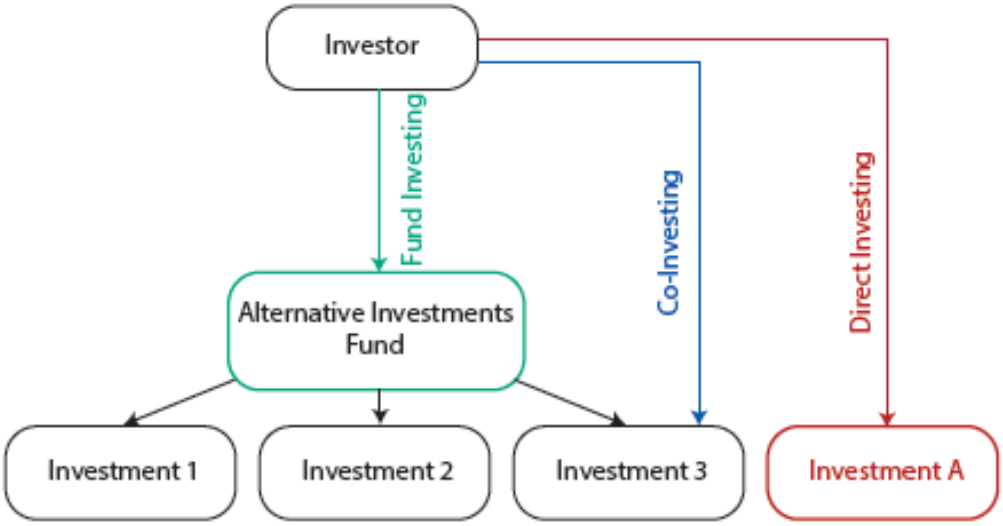
Alternative Investment Methods

The three methods of investing in alternative investments are:

- **Fund investing:** The investor contributes capital to a fund, and the fund makes investments on the investors' behalf, e.g., investments in a PE fund.
- **Co-investing:** The investor can make investments alongside a fund, e.g., investments in a portfolio company of a fund. In co-investing, the investor is able to invest both directly and indirectly in the same assets.
- **Direct investing:** The investor makes a direct investment in a company or project without the use of an intermediary, e.g., direct investments in infrastructure or real estate assets.

Exhibit 1 from the curriculum illustrates the three methods of investing in alternative investments:

© IFT. All rights reserved



Advantages and Disadvantages of Direct Investing, Co-Investing, and Fund Investing

The following table summaries the advantages and disadvantages of the different methods of investing.

	Advantages	Disadvantages
Fund investing	• Lower level of investor involvement as the fund manager provides investment services and expertise • Access to alternative investments without possessing a high degree of investment expertise • Lower minimum capital requirements	• Costly management and performance fees • Investor must conduct thorough due diligence when selecting the right fund because of the wide dispersion of fund manager returns • Investors less able to exit the investment as funds typically have lock-ups and other restrictions
Co-investing	• Investors can learn from the fund's process to become better at direct investing • Reduced management fees • Allows more active management of the portfolio compared with fund investing and allows for a deeper relationship with the manager	• Reduced control over the investment selection process compared with direct investing • May be subject to adverse selection bias • Requires more active involvement compared with fund investing, which can be challenging

© IFT. All rights reserved

Direct investing	• Avoids paying ongoing management fees to an external manager • Greatest amount of flexibility for the investor • Highest level of control over how the asset is managed	• Requires more investment expertise and a higher level of financial sophistication resulting in higher internal investment costs • Less access to a fund's ready diversification benefits or the fund manager's sourcing network • Requires more complex due diligence because of the absence of a fund manager • Higher minimum capital requirements

4. Alternative Investment Structures

Partnership Structures

The most common structure for many alternative investments is a partnership. It consists of two entities:

- **General partner (GP)**: The fund manager is the general partner (GP). The GP is responsible for managing the fund and making investment decisions. The GP theoretically bears unlimited liability for anything that goes wrong.
- **Limited partners (LP):** LPs are outside investors who provide capital to the fund in return for a fractional partnership in the fund. They bear the risk associated with investments. LPs usually play a passive role and are not involved with the management of the fund.

The partnership between the GP and LPs is governed by a **limited partnership agreement (LPA).** It is a legal document that outlines the rules of the partnership and establishes the framework for the fund's operations.

In addition to LPAs, **side letters** may also be negotiated, which are additional terms between the GP and certain LPs that exist outside the LPA (which is a common document for all LPs).

Other structures may be adopted for specific alternative investments. For example:

- Infrastructure investors often enter into public-private partnerships, which are agreements between the public sector and private sector.
- Real estate direct investing often involves a joint venture structure.
- Real estate fund investing often involves a unitholder structure.

© IFT. All rights reserved

Compensation Structures

The general partner typically receives a **management fee** based on assets under management (commonly used for hedge funds) or committed capital (commonly used for private equity). Management fee typically ranges from 1% to 2%.

Apart from the management fee, the GP also receives a **performance fee** (also called incentive fee or carried interest) based on realized profits. Performance fees are designed to reward GPs for good performance. A common fee structure is 2 and 20 which means 2% management fee and 20% performance fee.

Generally, the performance fee is paid only if the returns exceed a **hurdle rate** (also called a preferred rate). A hurdle rate of 8% is typically used.

- Hard hurdle rate: The GP earns fees on annual returns in excess of the hurdle rate.
- Soft hurdle rate: The GP earns fees on the entire annual gross return as long as the set hurdle is exceeded.

Common Investment Clauses, Provisions, and Contingencies

Common investment clauses, provisions and contingencies specified in the LPA include:

Catch-up clause: A catch-up clause allows the GP to receive 100% of the distributions above the hurdle rate until he receives 20% of the profits generated, and then every excess dollar is split 80/20 between the LPs and GP. This clause is meant to make the manager whole so that their incentive fee is a function of the total return and not solely on the return in excess of the hurdle rate.

Example:

Assume that the GP has earned an 18% IRR on an investment, the hurdle rate is 8%, and the partnership agreement includes a catch-up clause.

In this case the distribution would be as follows:

- The LPs would receive the entirety of the first 8% profit.
- The GP would receive the entirety of the next 2% profit—because 2% out of 10% amounts to 20% of the profits accounted for so far.
- The remaining 8% would be split 80/20 between the LPs and the GP

Thus, the GP effectively earns: 18% x 20% = 3.6% and the LP effectively earns 18% x 80% = 14.4%.

In the absence of a catch-up clause the distributions would have been:

- The LPs would still receive the entirety of the first 8% profit.
- The remaining 10% would be split 80/20 between the LPs and GP.

Thus, in this case the GP effectively earns a lower return of (18% - 8%) x 20% = 2.0%

© IFT. All rights reserved

High water mark: In some cases, the incentive fee is paid only if the fund has crossed the high-water mark. A high-water mark is the highest value net of fees (or the highest cumulative return) reported by the fund so far for each of its investors. This is to ensure investors do not pay twice for the same performance.

Waterfall: The waterfall defines the way in which cash distributions will be allocated between the GP and the LPs. In most waterfalls, a GP receives a disproportionately larger share of the total profits relative to their initial investment. This is typically done to incentivize GPs to maximize profitability.

There are two types of waterfalls:

- Whole-of-fund (or European) waterfalls: As deals are exited, all distributions go to the LPs first. The GP does not participate in any profits until the LPs receive their initial investment and the hurdle rate has been met.
- Deal-by-deal (or American) waterfalls: Performance fees are collected on a per-deal basis. This is more advantageous for a GP as he can get paid before LPs receive both their initial investment and their preferred rate of return on the entire fund.

Clawback: A clawback provision allows LPs to reclaim a part of the GP's performance fee. For example, if a fund makes profitable exits in early years, but the subsequent exits are less profitable, then the GP has to pay back profits to ensure that the profit split is in line with the fund prospectus.

© IFT. All rights reserved

Summary

LO: Describe features and categories of alternative investments.

Traditional investments refer to long-only positions in stocks, bonds, and cash. All other investments are classified as alternative investments.

Alternative investments can be divided into three main categories:

- Private capital – Includes private equity and private debt
- Real assets – Includes real estate, infrastructure, natural resources and others
- Hedge funds

Features that distinguish alternative investments from traditional investments are:

- Need for specialized knowledge: For example, within private equity, you have leveraged buyout and venture capital. There are managers who focus only on leveraged buyouts within private equity.
- Relatively low correlation with traditional investments. But correlation may increase during times of financial crisis.
- Illiquidity, long investment time horizons, and large capital outlays

LO: Compare direct investment, co-investment, and fund investment methods for alternative investments.

The three methods of investing in alternative investments are:

- Fund investing: The investor contributes capital to a fund, and the fund makes investments on the investors' behalf, e.g., investments in a PE fund.
- Co-investing: The investor can make investments alongside a fund, e.g., investments in a portfolio company of a fund.
- Direct investing: The investor makes a direct investment in a company or project without the use of an intermediary, e.g., direct investments in infrastructure or real estate assets.

LO: Describe investment ownership and compensation structures commonly used in alternative investments.

The most common structure for many alternative investments is a partnership. It consists of two entities: General partner (GP) who is responsible for managing the fund and making investment decisions, and limited partners (LPs) who provide capital to the fund in return for a fractional partnership in the fund.

The general partner typically receives a management fee based on assets under management (commonly used for hedge funds) or committed capital (commonly used for private equity). Apart from the management fee, the GP also receives a performance fee (also called

© IFT. All rights reserved

incentive fee or carried interest) based on realized profits. Generally, the performance fee is paid only if the returns exceed a hurdle rate.

© IFT. All rights reserved

LM02 Alternative Investment Performance and Returns

1. Introduction....2
2. Alternative Investment Performance....2
3. Alternative Investment Returns....6
Summary....10

This document should be read in conjunction with the corresponding learning module in the 2024 Level I CFA® Program curriculum. Some of the graphs, charts, tables, examples, and figures are copyright 2023, CFA Institute. Reproduced and republished with permission from CFA Institute. All rights reserved.

Required disclaimer: CFA Institute does not endorse, promote, or warrant the accuracy or quality of the products or services offered by IFT. CFA Institute, CFA®, and Chartered Financial Analyst® are trademarks owned by CFA Institute.

Version 1.0

© IFT. All rights reserved

1. Introduction

This learning module covers:

- Issues in performance appraisal of alternative investments
- Calculating fees and returns of alternative investments

2. Alternative Investment Performance

Alternative investments differ from traditional asset classes in the following ways:

- Longer time horizons
- Unique patterns of cash flows
- The use of leverage
- Illiquid positions
- More complex fee structures
- Different tax and accounting treatment
- Less normally distributed returns

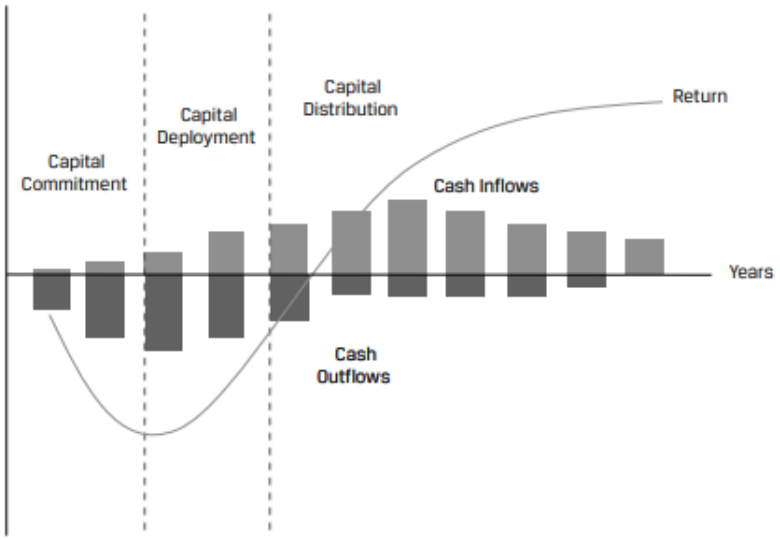
Due to these characteristics, it can be difficult to conduct performance appraisal on alternative investments.

When evaluating alternative investments, four factors should be considered:

- the life cycle phase of the investment
- the amount of borrowed funds used to maintain the market position
- the valuation of the assets
- the fee structure of the fund

Investment Life Cycle

Life cycle phases of various alternative investments generally fall into three distinct periods, as shown in Exhibit 1 from the curriculum.



© IFT. All rights reserved

Capital commitment: In this phase, managers identify and select appropriate investments with either an immediate or a delayed commitment of capital (known as a capital call). Returns are typically negative during this phase because fees and expenses are incurred immediately prior to capital deployment, and assets may generate little or no income during this period.

Capital deployment: In this phase, managers deploy funds to:

- construct or make property improvements (in case of real estate or infrastructure fund)
- incur expenses in the turnaround of a mature company (in case of private equity)
- initiate operations for a startup (in case of a venture capital)

Cash outflows typically exceed cash inflows. Management fees further reduce returns.

Capital distribution: If the property improvements/turnaround strategy/ or startup phase is successful, the underlying assets will appreciate in price and/or generate income exceeding costs. The fund can realize substantial capital gains from liquidating or exiting its investments.

Private Equity and Real Estate Performance Evaluation

Private equity and real estate investments often display a *J-curve effect* – initial decline followed by strong growth over the long term. This is because both private equity and real estate require significant initial cash outlays, and the investments take some time to turn profitable. Therefore, it is inappropriate to use short-term performance measures for private equity and real estate. Instead, the following measures are commonly used.

Performance measures for private equity:

The *IRR calculation* is frequently used to evaluate private equity investments. However, the determination of an IRR involves certain assumptions about a financing rate to use for outgoing cash flows (typically a weighted average cost of capital) and a reinvestment rate assumption to make on incoming cash flows (which must be assumed and may or may not actually be earned).

To overcome this complexity, the *multiple of invested capital (MOIC),* or money multiple is frequently used. It simply measures the total value of all distributions and residual asset values relative to an initial total investment.

> MOIC = (Realized value of investment + Unrealized value of investment)/(Total amount of invested capital).

Although simple to calculate, a major drawback of MOIC is that it ignores the timing of the cash flows.

© IFT. All rights reserved

Example:

(*This is Q#3 from the Question Set of the curriculum.*)

A private equity closes a fund with a capital commitment of €750 million. It has a capital call of €500 million initially and another €250 million at the end of Year 1. The management fee is 2% per annum. At the end of Year 5, a total of €1.0 billion is distributed to its investors, and the fund is left with €500 million in asset value. Calculate the fund's MOIC.

Solution:

MOIC = (Realized value of investment + Unrealized value of investment)/Total amount of invested capital, where: invested capital equals total paid-in capital less management fees and fund expenses. MOIC is different from the IRR measure because it ignores the timing of cash flows.

Total paid-in capital = 500 + 250 = 750.

Total management fee for 5 years = 750 × 0.02 × 5 = 75.

Total invested capital = 750 − 75 = 675.

MOIC = (1,000 + 500)/675 ≈ 2.2×.

Performance measures for real estate:

The *cap rate* is often used to evaluate real estate investments. It is calculated as the annual rent actually being earned divided by the price originally paid for the property.

Use of Borrowed Funds

Managers may use borrowed funds to increase investment returns. Leverage allows managers to take a market position that is larger than the capital committed.

Consider a cash investment V_c with a periodic rate of return r. If we assume an investor is able to borrow at a periodic rate of r_b to increase the size of its investment by borrowed funds of V_b, we can calculate a simple leveraged rate of return r_L for the period as follows:

$$r_L = \text{Leveraged portfolio return/Cash position} = [r \times (V_c+V_b) - (V_b \times r_b)]/V_c$$

This equation can be rearranged as:

$$r_L = r + V_b/V_c(r - r_b)$$

Hedge funds often use leverage to enhance returns. To lever their portfolio hedge funds use derivatives or borrow capital from prime brokers. Hedge funds must deposit cash or other collateral into a margin account with the prime broker, who then lends securities to the hedge fund. If the margin account falls below a certain threshold, a margin call is issued, and the hedge fund is required to put up additional collateral. This can magnify a hedge fund's losses because it may have to liquidate the losing position to meet the margin call.

© IFT. All rights reserved

Example:
(*This is based on Example 2 from the curriculum.*)

A hedge fund with a $100 million capital normally uses leverage to invest in a variety of equity-linked notes.

Scenario 1: Suppose the fund's underlying positions return 8%. If it could add leverage of USD50 million to the portfolio at a funding cost of 4%, what would have been the leveraged return?

The leveraged return can be calculated as follows:

$V_c = 100$; $V_b = 50$.

$r_L = 0.08 + (50/100)(0.08 - 0.04) = 10\%$

Scenario 2: Suppose the fund's underlying positions incur a loss of 2% instead of earning a gain. What would have been the leveraged return if the fund had borrowed USD50 million at 4%?

$V_c = 100$ and $V_b = 50$

$r_L = -0.02 + (50/100)(-0.02 - 0.04) = -5\%$.

Valuation

Alternative investment funds may be valued on a daily, weekly, monthly, and/or quarterly basis. The value of a fund depends on the value of underlying positions.

The price used for valuation depends on whether market prices are available and if the underlying position is liquid. When market prices are available, the fund decides what price to use. Common practice is to quote at the average of the bid and ask prices. A conservative approach is to use bid prices for long and ask prices for short.

GAAP accounting rules categorize fund investments into three buckets:

- Level 1: An exchange-traded, publicly traded price is available and is used for valuation purposes.
- Level 2: When such price is not available, outside broker quotes are used.
- Level 3: When broker quotes are not available or are unreliable, as a final recourse, assets are valued using internal models.

Level 3 assets values require additional scrutiny from investors. The models used should be appropriate and consistent. The values obtained may not reflect true liquidation values. Also, the returns may be smoothed and the volatility understated.

Fees

The after-fee results of alternative investments can vary significantly based on 'which' investor has invested 'when' in a particular fund.

For example:

© IFT. All rights reserved

- An investor may face significantly lower incentive fees if he invests more capital in a fund at an earlier phase or is willing to accept greater restrictions on redemptions.
- An investor who enters an alternative fund after it has experienced a sharp decline in value may incur performance fees if the fund rises, whereas an earlier investor who experienced the sharp decline in value from its peak may be exempt from such fees for the same period.

3. Alternative Investment Returns

Hedge funds have performance fees, and other customized, complex compensation arrangements that seek to align manager and investor incentives. These structures are designed to reward investors for early involvement, larger investments, and/or longer lockup periods.

Another factor that can magnify losses for hedge funds is redemption pressure. Redemptions usually occur when the hedge fund is performing poorly. Redemptions can force hedge fund managers to liquidate positions at disadvantageous prices.

To discourage redemptions:

- Hedge funds sometimes charge *redemption fees* (typically payable to the remaining investors) to offset the transaction costs for the remaining investors.
- Hedge funds use *notice periods* (investors need to inform the fund manager in advance before making a redemption) which provide the hedge fund manager an opportunity to liquidate positions in an orderly manner.
- Hedge funds use *lockup period* (time periods when investors cannot withdraw their capital) which provide the hedge fund manager sufficient time to implement his investment strategy.

Alternative Investment Returns

Hedge funds commonly use a "2 and 20" fee structure and fund of funds commonly use the "1 and 10" fee structure. However, many variations of the fee structure exist.

Analysts should be aware of any custom fee arrangements in place that will affect the calculation of fees and performance. These can include such arrangements such as:

- Fees based on liquidity terms and asset size: Hedge funds may provide a fee discount to investors who are willing to accept lower liquidity e.g., longer lockups. Similarly, hedge funds may provide a fee discount to larger investors. These terms are negotiated with individual investors via side letters, which are special amendments to the fund's LPA.
- Founder's share: To entice early participation in new hedge funds, managers often offer incentives known as founder's class shares. These shares have a lower fee structure e.g., "1.5 and 10" instead of "2 and 20" and are typically applicable to a certain cutoff threshold e.g., the first $100 million in assets.

© IFT. All rights reserved

- Either/or fees: A few large institutional investors have recently worked out a new fee model with some hedge fund managers. These managers agree either to charge a 1% management fee or to receive a 30% incentive fee above a mutually agreed-on hurdle rate, whichever is greater. The 1% management fee allows a fund to cover its expenses during down years and the 30% incentive fee incentivizes and rewards managers during up years.

Alternative Investment Return Calculations

Example: Incentive Fees Relative to Waterfall Types

A PE fund invests $10 million in Portfolio company A and $12 million in portfolio company B. Company A generates a $6 million profit, but Company B generates a $7 million loss. The time period for the gain and loss are the same. The manager's carried interest incentive fee is 20% of profits. Calculate the incentive fee under:

1. A European-style waterfall whole-of funds approach
2. An American-style waterfall deal-by-deal basis (assuming no clawback)

Solution to 1:

Overall, the fund lost money (+$6 million - $7 million = -$1 million) so under a European-style whole-of-fund waterfall, the manger will not receive any incentive fee

Solution to 2:

Under an American-style waterfall, the GP could still earn 20% x $6 million = $1.2 million as incentive fees on the profitable Company A deal.

Example: Fee and return calculations

Consider a hedge fund with an initial investment of 200 million; the fee structure is 2 and 20 and is based on year-end valuation. In year 1, the return is 30%.

1. What is the total fee if management fee and incentive fee are calculated independently? What is the investor's effective return?
2. What is the total fee if the incentive fee is calculated after deducting the management fee? Investor's net return?
3. If there is a hurdle rate of 5% and fees are based on returns of in excess of 5%, what is the total fee? What is the investor's net return?
4. In the second year, the fund declines to 220 million. Assume that management fee and incentive fee are calculated independently as indicated in Part 1, but now a high-water mark is also used in fee calculations. What is the total fee? What is the investor's net return?
5. In the third year, the fund value increases to 256 million. What is the total fee and investor's net return?

Solution:

© IFT. All rights reserved

1. Initial investment grows to: 200 x 1.3 = $260 million.
Profit = $60 million.
Management fee: 0.02 x 260 = $5.2 million.
Incentive fee which is 20% of profit = 20% x 60 = $12 million.
Total fee = $5.2 million + $12 million = $17.2 million.
Investor's return = $\frac{260-17.2}{200} - 1 = 21.4\%$

2. Incentive fee after deducting management fee = 20% x (260 – 200 - 5.2) = 10.96.
Total fee = 5.2 + 10.96 = $16.16 million.
Investor's return = $\frac{260-16.16}{200} - 1 = 21.92\%$.
As you can see the return is better than Part 1 because incentive fee paid is relatively less here.

3. There is a hurdle rate of 5%. So, 200 x 0.05 = $10 million must be subtracted before incentive fees are paid.
Incentive fee = 0.2 x (260 – 200 - 5.2 - 10) = 8.96.
Total fee = 5.20 + 8.96 = $14.16 million.
Incentive fee is further reduced and the investor's return is enhanced.
Investor's return = $\frac{260-14.16}{200} - 1 = 22.92\%$.

4. Management fee = 0.02 x 220 = 4.4. To calculate the incentive fee, we need to determine whether the fund value has exceeded the high-water mark. The high-water mark was achieved at the end of Year 1. This value was 260 million – 17.2 million = 242.8 million. The incentive fee is 0 because the fund value is below the high-water mark. Hence the total fee = $4.4 million.
Investor's return = $\frac{220-4.4}{242.8} - 1 = -11.2\%$

5. Management fee = 256 x .02 = 5.12. Since $256 has exceeded high water mark of 242.8 million, an incentive fee would be paid. Incentive fee = (256 - 242.8) x 0.2 = 2.64. Total fee = 5.12 + 2.64 = 7.76 million.
Investor's net return = $\frac{256-7.76}{215.6} - 1 = 15.14\%$.

Example: Hedge fund versus fund of funds

An investor is contemplating investing £200 million in either the Hedge Fund (HF) or the Fund of Funds (FOF). FOF has a "1 and 10" fee structure and invests 10% of its assets under management in HF. HF has a standard "2 and 20" fee structure with no hurdle rate. Management fees are calculated on an annual basis on assets under management at the beginning of the year. Management fees and incentive fees are calculated independently. HF has a 25% return for the year before management and incentive fees.

1. Calculate the return to the investor of investing directly in HF.

© IFT. All rights reserved

2. Calculate the return to the investor of investing in FOF. Assume that the other investments in the FOF portfolio generate the same return before management fees as HF and have the same fee structure as HF.

Solution to 1:

HF has a profit before fees on a £200 million investment of £50 million (= 200 million × 25%). The management fee is £4 million (= 200 million × 2%) and the incentive fee is £10 million (= 50 million × 20%). The return to investor is 18% (= (50 – 4 – 10) / 200).

Solution to 2:

FOF earns an 18% return or £36 million profit after fees on £200 million invested with hedge funds. FOF charges the investor a management fee of £2 million (= 200 million × 1%) and an incentive fee of £3.6 million (= 36 million × 10%). The return to the investor is 15.2% (= (36 – 2 -3.6) / 200).

Relative Alternative Investment Returns and Survivorship Bias

Hedge fund index returns can be overstated due to survivorship, and backfill biases.

- Survivorship bias occurs when an index is composed of only surviving funds over a period of time, which tends to overstate the index returns.
- Backfill bias occurs when a new fund enters a database and historical returns of that fund are added (i.e., "backfilled"). Usually, funds that performed well are added which tends to overstate the index returns.

Example: Clawbacks Due to Return Timing Differences

A PE fund makes two investments for $5 million each in Company A and Company B. One year later Company A returns a $8 million profit. But two years later Company B turns out to be a complete bust and is worth zero.

The GP's carried interest is 20% of aggregate profits and there is a clawback provision. How much carried interest will the GP receive in year 1 and year 2.

Solution:

In year 1, the GP will receive a carried interest of 20% of $8 million = $1.6 million. This amount would typically be held in an escrow account for the benefit of the GP but not actually paid.

In year 2, the GP loses $5 million of the initial $8 million gain, so the aggregate profit is only $3 million. The carried interest payable is 20% x $3 million = $0.6 million. The GP has to return $1 million of the previously accrued incentive fee to the LPs because of the clawback provision.

© IFT. All rights reserved

Summary

LO: Describe the performance appraisal of alternative investments.

It can be difficult to conduct performance appraisal on alternative investments because these investments have asymmetric risk–return profiles, limited portfolio transparency, illiquidity, product complexity, and complex fee structures.

The IRR and MOIC calculations are frequently used to evaluate private equity investments. The cap rate is frequently used to evaluate real estate investments.

Leverage, illiquidity and redemption pressure pose special challenges while evaluating hedge funds' performance.

LO: Calculate and interpret alternative investment returns both before and after fees.

Alternative investment managers usually charge a management fee based on AUM and an incentive fee based on performance. However, analysts should also be aware of any custom fee arrangements in place that will affect the calculation of fees and performance. These can include such arrangements such as: fees based on liquidity terms and asset size, founder's share, and either/or fees.

It is difficult to generalize performance appraisal for these investments because returns may vary depending on how and when a particular investor invested in a particular vehicle.

Hedge fund index returns can be overstated due to survivorship, and backfill biases.

© IFT. All rights reserved

LM03 Investments in Private Capital - Equity and Debt

1. Introduction....2
2. Private Equity Investment Characteristics....2
3. Private Debt Investment Characteristics....6
4. Diversification Benefits of Private Capital....7
Summary....8

This document should be read in conjunction with the corresponding learning module in the 2024 Level I CFA® Program curriculum. Some of the graphs, charts, tables, examples, and figures are copyright 2023, CFA Institute. Reproduced and republished with permission from CFA Institute. All rights reserved.

Required disclaimer: CFA Institute does not endorse, promote, or warrant the accuracy or quality of the products or services offered by IFT. CFA Institute, CFA®, and Chartered Financial Analyst® are trademarks owned by CFA Institute.

Version 1.0

1. Introduction

This learning module covers:

- Features and investment characteristics of private equity
- Features and investment characteristics of private debt
- Diversification benefits of private capital

2. Private Equity Investment Characteristics

Private capital is a broad term for funding provided to companies that is not sourced from the public equity or debt markets.

Capital that is provided in the form of equity investments is called private equity, whereas capital that is provided as a loan or other form of debt is called private debt.

Private Equity: Description

Private equity means investing in private companies or public companies with the intent to take them private. The companies in which the private equity funds invests are called *portfolio companies* because they will become part of the private equity fund portfolio.

The three main categories of private equity are:

- Leveraged buyouts: Borrowed funds are used to buy an established company.
- Venture capital: Refers to investments in companies that have not been established yet.
- Growth capital: Refers to minority equity investments in mature companies that require funds for growth or expansion, restructuring, entering a new territory, an acquisition, etc.

Leveraged Buyouts

Leveraged buyout is an acquisition of an established public or private company with borrowed funds. If the target company is a public company, then after the acquisition, the company becomes private, i.e., the target company's equity is no longer publicly traded.

The acquisition is significantly financed through debt, hence the name leveraged buyout. LBOs capital structure consists of equity, bank debt, and high-yield bonds. The firm (GP) puts in some money of its own, raises a certain amount from LPs, and a substantial amount of money is borrowed in the form of debt to invest in companies.

For example, assume the GP invests in a target company that requires an investment of $100 million. In this, the GP invests $20 million of its money (equity), $70 million from bank debt, and the remaining $10 million is raised by issuing high-yield bonds.

There are three changes that happen to a company as a result of a leveraged buyout:

- An increase in financial leverage.
- Change in management or the way the company is run.

© IFT. All rights reserved

- If the target company is previously public, after the LBO it becomes private.

Why LBO?

- To improve the company's operations; to add value and eventually increase cash flows and profits.
- Leverage will enhance potential returns once the restructuring/growth strategy is complete and the company turns profitable. Debt is central to an LBO structure. Buyouts are rarely done entirely using equity.

There are two types of LBOs:

- Management buyouts (MBO): Current management team purchases and runs the company.
- Management buy-ins (MBI): Current management team is replaced and the acquirer team runs the company.

Venture Capital

Venture capital firms invest in private companies (portfolio companies) with significant growth potential. The time horizon is typically long-term. The distinction between VC and LBO is that the latter invests in mature companies, whereas VC invests in growing companies with a good business plan and strong prospects for future growth.

Other important points related to VCs are given below:

- Venture capitalists are actively involved in the companies they invest in.
- The rate of return expected depends on the stage the company is in when the investment happens.
- VC investing can take place at various stages

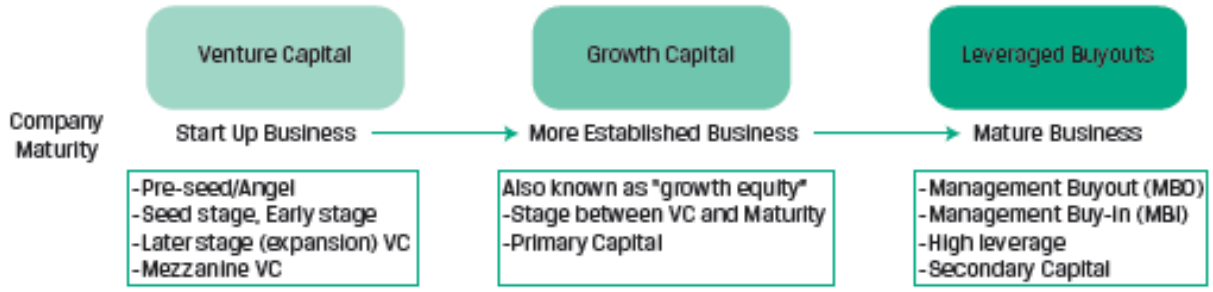
  Formative stage: Company is still being formed.
  - *Angel investing*: Financing provided at the idea stage.
  - *Seed stage financing*: Financing provided for product development and market research.
  - *Early stage*: Financing for companies moving towards operation, but before commercial production and sales. Fund to initiate commercial production and sales.

  Later stage financing: For expansion after commercial production and sales but before IPO.

  Mezzanine stage: Preparing to go public.

The following exhibit shows the growth stages of a company and the types of financing it may receive at each stage.

© IFT. All rights reserved



PIPE (Private Investment in public equity):

- A PIPE occurs when an institutional or accredited investor purchases stock from a public company at a discount to the market price.
- PIPEs save companies time and money while raising funds because they have less stringent regulatory requirements than public offerings.
- However, the lower cost of PIPE shares means less capital for the company, and their issuance effectively dilutes the stake of current stockholders.

Private Equity Exit Strategies

The goal of private equity is to improve new or underperforming businesses and exit them at high valuations. Typically, investments (target companies) are held for an average of 5 years. The holding period may be longer or shorter.

The three common exit strategies are:

- Trade sale: Selling the company to a competitor or any strategic buyer. It can be done through auction or private negotiation. For instance, if a PE firm (GP) invested in a small generic pharma company, it may sell it to large pharma firm after a few years.
- IPO: Company goes public, i.e., it sells all or some of its shares to public investors.
- Special purpose acquisition company (SPAC): A SPAC is a shell company, often called a "blank check" company, because it exists solely for the purpose of acquiring an unspecified private company sometime in the future. SPACs raise capital through IPOs and deposit the proceeds in a trust account. They have a finite time (e.g. 24 months) to complete a deal; otherwise, the proceeds are returned back to the investors.

Exhibit 3 from the curriculum lists the pros and cons of these three strategies.

© IFT. All rights reserved

Strategy	Advantage	Disadvantage
Trade Sale	1. Immediate cash exit 2. Higher price from synergy-seeking strategic buyers 3. Fast and simple execution 4. Streamlined process on transaction cost, disclosure, and confidentiality from dealing with only one party	1. Potential management opposition 2. Limited set of buyers 3. Reduced financial appeal to employees due to forgone monetization of ownership stakes/options
IPO	1. Highest potential share price 2. Likeliest management approval 3. Notoriety to private equity sponsor' 4. Sharing in potential share price appreciation from ongoing ownership stake	1. High transaction costs 2. Long lead time 3. Stock market volatility creating value uncertainty 4. Onerous disclosure 5. Potential lockup period freezing capital committed to deal 6. Suitable mainly for large and fast-growing companies
SPAC	1. Extended disclosure time and ability to provide forward guidance to develop investor interest 2. Fixed valuation with lower share price volatility 3. Transaction structure flexibility 4. Involvement of high-profile, seasoned sponsors and their investor networks	1. Potential higher capital costs of dilution, warrants, and fees 2. Divergence between announced and true equity value due to dilution 3. Deal and capital risk of potential redemptions 4. Prolonged post-merger stockholder overhang and churn

Other exit strategies include:

- Recapitalization: The PE firm increases leverage or introduces it to the portfolio company and pays itself a dividend out of the new capital structure. Not a true exit strategy, as the PE firm still maintains control, but it does allow the PE firm to extract money from the company.
- Secondary sale: Assume you are a VC firm that focuses on early-stage companies. You may sell the portfolio company later to another private equity firm that focuses on later stage companies.
- Write off/liquidation: This is a worst-case scenario when the investment has not gone as planned. The company's prospects do not look promising, so the VC firm sells the assets or writes it off to focus on other projects.

Risk–Return from Private Equity Investments

Private equity may provide higher return opportunities relative to traditional investments. Some of its benefits include the following:

- Access to private companies.
- Ability to actively manage and improve portfolio companies.

© IFT. All rights reserved

- Easy to use leverage.

However, the higher return is often associated with higher illiquidity and leverage risks.

3. Private Debt Investment Characteristics

Private Debt Categories

Private debt refers to various forms of debt provided by investors to private entities.

Key private debt categories include:

- Direct lending: Debt capital is provided at higher interest rates, directly to entities that require capital, but are unable to get capital from traditional bank lenders. Lenders subsequently receive interest, the original principal, and possibly other payments in exchange for their investment.

  In direct lending, many firms may also provide debt in the form of a *leveraged loan*, a loan that is itself levered. Private debt firms that invest in leveraged loans first borrow money to finance the debt and then extend it to another borrower. Through leverage, a private debt fund can enhance the return on its loan portfolio.

- Mezzanine debt: Refers to private credit that is subordinated to senior secured debt but is senior to equity in the borrower's capital structure. Because of the higher risk, investors commonly demand a higher interest rate and may also require options for equity participation.

- Venture debt: Debt funding provided to start-up or early-stage companies that may be generating little or negative cash flow. Entrepreneurs may seek venture debt as a way to access funds without further diluting shareholder ownership in their companies. Similar to mezzanine debt, venture debt may contain additional feature that compensate investors for the increased risk.

- Distressed debt: Refers to buying debt of mature companies with financial difficulty such a bankruptcy proceeding. Investors seek companies with a temporary cash-flow problem but a good business plan. They may also get actively involved in the management of the company and help turn it around.

- Unitranche debt: It consists of a hybrid or blended loan structure combining different tranches of secured and unsecured debt into a single loan with a single, blended interest rate. The interest rate will generally fall between the interest rates demanded on secured and unsecured debt.

Risk–Return of Private Debt

Private debt, like private equity, can be arranged on a direct or indirect basis, with funds deployed directly from an investor or through a fund over the corporate life cycle.

© IFT. All rights reserved

Investors receive interest payments and the return of principal at the end of a specified term, with debt typically secured and protected by covenants.

Private debt investments can provide a higher return as compared to traditional bonds. However, this higher return if often connected to higher levels of risk.

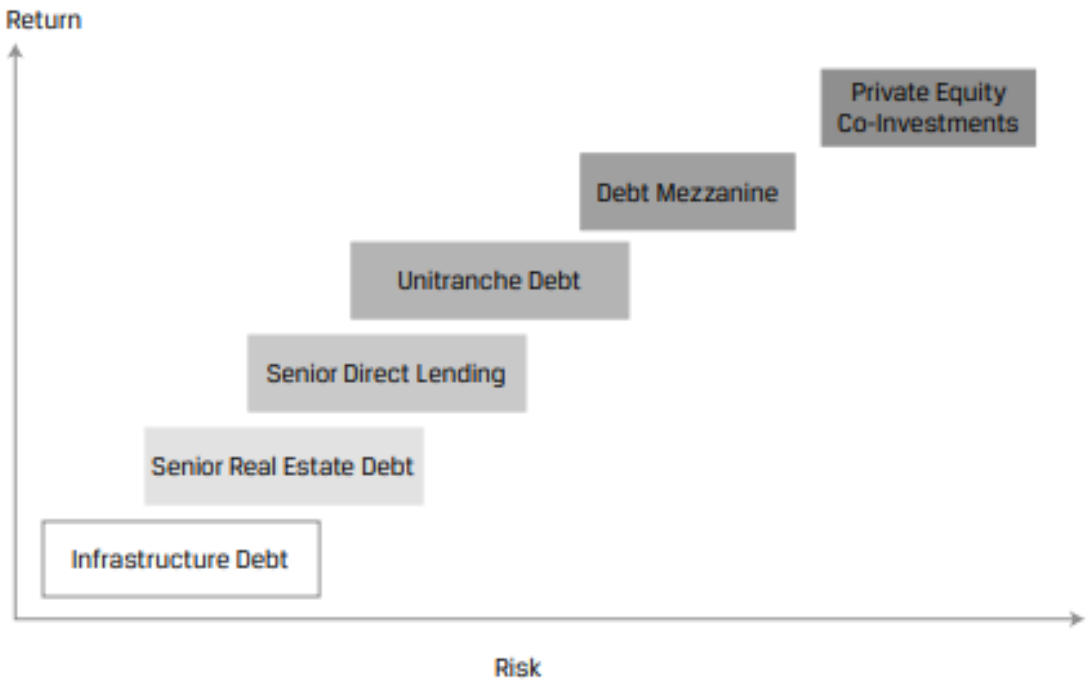
4. Diversification Benefits of Private Capital

Due to illiquidity and concentration risk, as well as the often-greater uncertainties of both their underlying businesses and the means to hedge away their risks, private debt and equity differ from their public counterparts in terms of risks and performance.

Vintage year:

- A fundamental timing characteristic for private capital is its vintage year (year of launch). The valuation and economic environment at the fund's inception can have a significant impact on realized results over the fund's lifespan.
- To offset the potentially adverse impact of an ill-timed fund launch at an unfavorable stage of the business cycle, investors can diversify exposure across fund vintage years.

Investments in private capital vary in terms of risk and return across the corporate capital structure hierarchy. Exhibit 6 from the curriculum plots the various private equity and private capital categories by their risk and return levels.



Investments in private capital funds can add a moderate diversification benefit to a portfolio of publicly traded stocks and bonds.

© IFT. All rights reserved

Summary

LO: Explain features of private equity and its investment characteristics.

Private equity means investing in private companies or public companies with the intent to take them private. The three main categories of private equity are: leveraged buyouts, venture capital, and growth capital. The main exit strategies are: trade sale, IPO, and SPAC.

LO: Explain features of private debt and its investment characteristics.

Private debt refers to various forms of debt provided by investors to private entities. Key private debt categories include: direct lending, mezzanine debt, venture debt, and distressed debt.

LO: Describe the diversification benefits that private capital can provide.

A fundamental timing characteristic for private capital is its vintage year (year of launch). The valuation and economic environment at the fund's inception can have a significant impact on realized results over the fund's lifespan.

Investments in private capital funds can add a moderate diversification benefit to a portfolio of publicly traded stocks and bonds.

© IFT. All rights reserved

LM04 Real Estate and Infrastructure

1. Introduction....2
2. Real Estate Features....2
3. Real Estate Investment Characteristics....5
4. Infrastructure Investment Features....5
5. Infrastructure Investment Characteristics....7
Summary....9

This document should be read in conjunction with the corresponding learning module in the 2024 Level I CFA® Program curriculum. Some of the graphs, charts, tables, examples, and figures are copyright 2023, CFA Institute. Reproduced and republished with permission from CFA Institute. All rights reserved.

Required disclaimer: CFA Institute does not endorse, promote, or warrant the accuracy or quality of the products or services offered by IFT. CFA Institute, CFA®, and Chartered Financial Analyst® are trademarks owned by CFA Institute.

Version 1.0

1. Introduction

This learning module covers:

- Features and investment characteristics of real estate
- Features and investment characteristics of infrastructure

2. Real Estate Features

Real estate has two major sectors:

- Residential: Includes individual single-family detached homes and multi-family attached units owned by the residents. Residential real estate is the largest sector, making up some 75% of the market globally.
- Commercial: Commercial real estate primarily includes office buildings, shopping centers, and warehouses. When residential real estate properties (described above) are owned with the intent to rent, they are classified as commercial real estate.

Real Estate Investments

Exhibit 1 from the curriculum lists the main features of residential and commercial real estate.

	Residential real estate	Commercial real estate
Typical property	Owner-occupied, single residences; single-family residential property	Residential properties owned for lease or rental Office, retail, industrial, warehouse, hospitality, and mixed-use properties
Source of equity	Owners	Privately held by owners Publicly held through investors
Source of debt	Directly: Lenders (banks) through residential mortgages Indirectly: Investors in MBS that package residential mortgages	Directly: Lenders (banks) through commercial mortgages Indirectly: Investors in MBS that package commercial mortgages
Source of return to investors	Enjoyment of the property Price, or capital, appreciation	Income, or cash flow, generated by the property Price, or capital, appreciation

Real estate is different from other asset classes in several ways:

- Indivisibility – requires large capital investments
- Unique characteristics (no two properties are identical).
- There are various types of real estate investment alternatives available, ranging from

© IFT. All rights reserved

relatively liquid investments in stable, income-producing properties to illiquid investments over a long development life cycle spanning the purchase, construction/upgrade, occupancy, and sales phases.
- It can be difficult to diversify
- Indexes replicating the performance of real estate are not directly investable.

Also, the price discovery process in private real estate markets is opaque:
- Historical prices may not reflect current market conditions.
- Transactions are costly and time consuming.
- In some markets transaction activity may be limited due to supply or demand conditions.

Because of these unique features, real estate markets are typically fragmented. The local demand and supply conditions determine the value of a property, and local markets can be very different from national or global markets.

Real Estate Investment Structures

Real estate investing can be categorized along two dimensions: public/private markets and debt/equity based. Exhibit 2 presents the four quadrants for the basic forms of real estate investments with examples:

	Debt	Equity
Private	Mortgage debt Construction loans Mezzanine debt	**Direct ownership** Sole ownership Joint ventures Limited partnerships **Indirect ownership** Real estate funds Private REITs
Public	MBS/CMBS/CMOs Covered bonds Mortgage REITs Mortgage ETFs	**Publicly traded shares** Construction Operating Development Public REITs UCITS/Mutual funds/ETFs

Equity-based investments represent ownership of real estate properties. Ownership can be through sole ownership, joint ventures, real estate limited partnerships, etc. A variation of equity-based investments is leveraged ownership: Assume a building costs $10 million, and you put $3 million of your money and borrow $7 million. This is called leveraged ownership. That is, leveraged ownership is where a property is obtained through equity and mortgage financing.

© IFT. All rights reserved

If you are investing in a **debt-based** real estate investment, it means you are lending money to a purchaser of real estate. A classic example is a mortgage loan. This is considered a real estate investment because the value of the mortgage loan is related to the value of the underlying property.

There can be many variations within the basic forms:

- Direct real estate investing: Involves purchasing a property and originating debt for one's own account. The major advantages are: control, and tax benefits. The major disadvantages are: extensive time and expertise required to manage the property, the large capital requirements, and highly concentrated portfolios.
- Indirect real estate investing: Pooled investment vehicles are used to access the underlying real estate assets. The vehicles can be public or private, such as limited partnerships, mutual funds, corporate shares, REITs, and ETFs.
- Mortgages: Represent passive investments in which the lender can expect to receive a predefined stream of payments over the life of the mortgage.
- Private fund investing styles: Most real estate private equity funds are structured as infinite-life open-end funds, which allow investors to contribute or redeem capital throughout the life of the fund.
- REITs: REITs combine the features of mutual funds and real estate. An REIT is a company that owns income-producing real estate assets. In REITs, average investors pool their capital to invest (take ownership) in several large-scale, diversified income-generating real estate properties. The REIT issues shares, where each share represents a percentage ownership in the underlying property. The income generated is paid as a dividend to the shareholders.

  The main advantage of the REIT structure is that it avoids double corporate taxation. Normal corporations pay taxes on income, and then the dividend paid from the after-tax earnings are taxed again at the shareholder's personal tax rate. REITs can avoid corporate income taxes by distributing 90% - 100% of their rental income as dividends. Another advantage is that REITs are more transparent than private real estate markets.

  The value of the REIT shares is based on the dividend. REIT shares often trade publicly on exchanges. It is a way for individual investors to earn a share of the income from commercial properties (office buildings, warehouses, and shopping malls) without buying them. Risk and return of REITs vary based on the types of properties they invest in.

  Equity REITs invest in properties outright or through partnerships and joint ventures. The business strategy for equity REITs is simple: Maximize property occupancy rates and rents while minimizing ongoing operating and maintenance expenses to maximize

© IFT. All rights reserved

cash income and dividends. Mortgage REITs invest in real estate debt, typically MBS. Hybrid REITs invest in both real estate debt and equity.

3. Real Estate Investment Characteristics

The key reasons for investing in real estate are:

- Potential for competitive long-term returns (income and capital appreciation).
- Rent for long-term leases will lessen the impact of economic shocks.
- Diversification because of low correlation with other asset classes such as stocks, bonds.
- Inflation hedge.

Source of Returns

The return on real estate investments comes from income or asset appreciation or a combination of both. More than half the returns earned by commercial real estate investors comes from income, and it is a more consistent source of return throughout an economic cycle, as compared to capital appreciation.

Real estate investments can be similar to bond investments in that they generate stable, predictable, lower-risk cash flows from leases that are comparable to bond coupon payments.

Real estate investments are similar to equity investments in that they provide speculative returns based on the price appreciation of the real estate asset.

Real Estate Investment Diversification Benefits

Many investors prefer real estate for its ability to provide high, steady current income. Real estate also has low correlation with other asset classes and thus provides diversification benefits. However, there are periods when equity REIT correlations with other securities are high, and their correlations are highest during steep market downturns.

4. Infrastructure Investment Features

The assets underlying infrastructure investments are real, capital intensive, and long-lived. These assets are intended for public use, and they provide essential services e.g., airports, health care facilities, and power plants.

Infrastructure Investments

Like real estate, investment in infrastructure involves acquiring unique, illiquid assets with distinct locations, features, and uses. The returns can come from income and capital appreciation.

Rather than rentals, infrastructure cash flows arise from contractual payments such as:

- Availability payments: Payments received to make the facility available.
- Usage-based payments: Tolls and fees for using the facility.

© IFT. All rights reserved

- "Take-or-pay" arrangements: This obligates buyers to pay a minimum purchase price to sellers for a pre-agreed volume.

Infrastructure assets were primarily owned, financed, and operated by the government. Of late, they are financed privately through the use of public-private partnerships (PPPs). A public–private partnership is typically defined as a long-term contractual relationship between the public and private sectors for the purpose of having the private sector deliver a project or service traditionally provided by the public sector.

Categories of Infrastructure Investments

Infrastructure investments may be categorized based on: (1) underlying assets, and (2) stage of development of the underlying assets.

Infrastructure investments based on underlying assets: They can be classified into economic and social infrastructure assets.

- Economic infrastructure assets: These include transportation, communication, and social utility assets that are needed to support economic activity. Examples of transportation assets are roads, airports, bridges, tunnels, ports, etc. Examples of utility assets are assets used to transmit and distribute gas, electricity, generate power, etc. Examples of communication assets are assets that are used to broadcast information.
- Social infrastructure assets: These are assets required for the benefit of the society such as educational and healthcare facilities.

Infrastructure investments based on the stage of development of the underlying assets: They can be classified into greenfield, brownfield and secondary-stage investments.

- Greenfield investments develop new assets and new infrastructure. The objective may be to construct and sell the assets to the government, or to hold and operate the assets.
- Brownfield investments expand existing facilities and may involve privatization of public assets. They have a shorter investment period and immediate cash flows. The assets may also have a financial and operating history.
- Secondary-stage investments invest in existing infrastructure facilities or fully operational assets that do not need further expansion and investments. They generate immediate cash flows.

Forms of Infrastructure Investments

Investors may invest either directly or indirectly in infrastructure investments. The investment form affects the liquidity and the income and cash flows to the investor.

The advantages of investing directly in infrastructure are that investors have a control over the asset and can capture the full value of the asset. But the downside of a large investment is

© IFT. All rights reserved

that it results in concentration and liquidity risks.

Most investors invest indirectly. Some forms of indirect investments include:

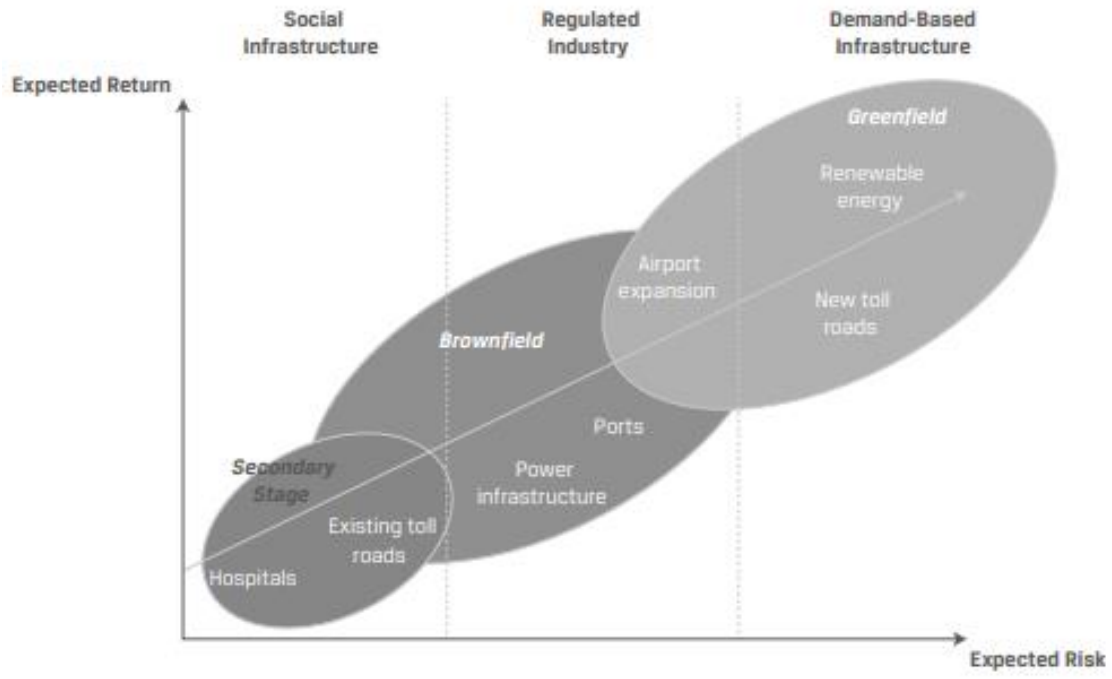
- investment in an infrastructure fund
- infrastructure ETFs
- shares of companies

Investing in publicly traded infrastructure companies offer the benefit of liquidity. Publicly traded infrastructure securities also have a reasonable fee structure, transparent governance, and provide the benefit of diversification. Master limited partnerships (MLPs) are pass-through entities similar to REITs and are listed on exchanges.

Infrastructure debt financing can take the form of both private and publicly traded debt. The terms are typically flexible to accommodate periods of zero cash flow and long development or investment horizons. Because of the stable underlying nature of cash flows, infrastructure debt has lower default rates and higher recovery rates than comparable fixed-income instruments, and it is less volatile over the economic cycle.

5. Infrastructure Investment Characteristics

Exhibit 8 from the curriculum depicts the investment characteristics of different types of infrastructure investments.



Greenfield infrastructure investments have the highest expected return and the highest expected risk, while secondary stage investments have the lowest expected return and the lowest expected risk.

© IFT. All rights reserved

Also, investments in basic social services infrastructure (e.g., hospitals) or existing regulated industries (e.g., Power grids) typically involves lower risk and lower expected return. Whereas, demand-based infrastructure (e.g., new toll roads) are built on projections of future economic growth and are riskier.

Infrastructure Diversification Benefits

Some of the advantages to investors from investing in infrastructure are as follows:

- a steady income stream
- potential for capital appreciation
- diversification because of low correlation of infrastructure assets to traditional investments
- protection against inflation
- match the long-term liability structure of some investors such as pension funds

© IFT. All rights reserved

Summary

LO: Explain features and characteristics of real estate.

Real estate includes two major sectors: residential and commercial.

Unique features of real estate are: heterogeneity, fragmentation, challenges in price discovery, costly and time-consuming transactions.

LO: Explain the investment characteristics of real estate investments.

Real estate investing can be categorized along two dimensions: public/private markets and debt/equity based.

The return on real estate investments comes from income or asset appreciation or a combination of both.

Real estate investments can be similar to bond investments in that they generate stable, predictable, lower-risk cash flows from leases that are comparable to bond coupon payments. Real estate investments are similar to equity investments in that they provide speculative returns based on the price appreciation of the real estate asset.

Real estate provides diversification benefits because of their low correlation with other asset classes. However, during steep market downturns, equity REIT correlations with market benchmarks can increase.

LO: Explain features and characteristics of infrastructure.

The assets underlying infrastructure investments are real, capital intensive, and long-lived. These assets are intended for public use, and they provide essential services.

Infrastructure investments can be categorized based on:

- underlying assets as either economic or social
- stage of development of underlying assets as greenfield, brownfield, or secondary stage.

Infrastructure assets were primarily owned, financed, and operated by the government. Of late, they are financed privately through the use of public-private partnerships (PPPs).

LO: Explain the investment characteristics of infrastructure investments.

Greenfield infrastructure investments have the highest expected return and the highest expected risk, while secondary stage investments have the lowest expected return and the lowest expected risk.

Some of the advantages to investors from investing in infrastructure are as follows:

- a steady income stream
- potential for capital appreciation

© IFT. All rights reserved

- diversification because of low correlation of infrastructure assets to traditional investments
- protection against inflation
- match the long-term liability structure of some investors such as pension funds

© IFT. All rights reserved

LM05 Natural Resources

1. Introduction....2
2. Natural Resources Investment Features....2
3. Commodity Investment Forms....5
4. Natural Resource Investment Risk, Return, and Diversification....6
Summary....8

This document should be read in conjunction with the corresponding learning module in the 2024 Level I CFA® Program curriculum. Some of the graphs, charts, tables, examples, and figures are copyright 2023, CFA Institute. Reproduced and republished with permission from CFA Institute. All rights reserved.

Required disclaimer: CFA Institute does not endorse, promote, or warrant the accuracy or quality of the products or services offered by IFT. CFA Institute, CFA®, and Chartered Financial Analyst® are trademarks owned by CFA Institute.

Version 1.0

1. Introduction

Natural resources include:

- **Commodities:** Can be further classified into:
  - Hard: Commodities that are mined e.g., copper, gold, silver; and commodities that are extracted e.g., crude oil, natural gas.
  - Soft: Commodities that are grown over a period of time e.g., grains, livestock, and cash crops like coffee.
- **Farm land:**
  - Investments in land used for the cultivation of crops or livestock.
  - Income can be generated from the growth, harvest and sale of crops or livestock; or by leasing the land back to farmers.
- **Timberland:**
  - Investments in natural forests or managed tree plantations.
  - The return comes from the sale of trees, wood, and other timber products.

Up to about 20 years ago, investors looking for exposure to natural resources invested mainly via financial instruments (stocks and bonds). Instead of investing in the physical land and the products that come from it, investors focused on the companies that produced natural resources. Nowadays, however, due to the wide variety of direct investments available (ETFs, limited partnerships, REITS, swaps, and futures), investors typically participate in these assets directly.

This learning module covers:

- Features and investment characteristics of raw land, timberland, and farmland
- Features and investment characteristics of commodities
- Sources of risk, return, and diversification benefits of natural resource investments

2. Natural Resources Investment Features

Timberland

Timberland provides an income stream through the sale of trees, wood, and other timber products. Timberland can be thought of as both a factory and a warehouse. The trees can be easily stored by simply not harvesting them. The trees can be harvested based on the price: more harvest when prices are up and delayed harvest when prices are down.

The three return drivers for timberland investments include: biological growth, change in prices of lumber (cut wood), and underlying land price change.

Additionally, since trees consume carbon as part of their life cycle, timberland considered a sustainable investment that mitigates climate-related risks.

© IFT. All rights reserved

Farmland

Farmland is perceived to provide a hedge against inflation. Two types of farm crops include raw crops that are planted and harvested, and permanent crops that grow on trees. Like timberland, farmland also provides an income component related to harvest quantities and agricultural commodity prices. However, it does not provide production flexibility, as farm products must be harvested when ripe.

Similar to timber land, the return drivers for farmland are: harvested quantities, commodity prices, and land price appreciation.

Farmland is also considered a sustainable investment that mitigates climate-related risks.

Land Investments vs. Real Estate

Farmland, timberland, and raw land are similar to real estate investments in that they are unique, illiquid assets with distinct geographic location and features. However, there are also some notable differences:

- Unlike real estate, there is no focus on the physical development of the land. Rather than development, the quality of the soil and climate features matter more.
- The location of the land is important. The closer it is to transportation hubs and markets, the higher the price. This is because transportation expenses can be a significant component of the price of the products paid by the customers of timberland and farmland.

To invest in raw land, timberland, and farmland, investors need specialized knowledge and understanding of the specifics of the natural resources. For example, to invest directly in timberland one needs forest expertise to manage a forest over its life cycle. Many institutional investors do not have this expertise, and rely on timberland investment management organizations (TIMOs)- TIMOs are entities that use their forest investment expertise to analyze and acquire suitable timberland holdings on behalf of institutional investors.

Exhibit 1 from the curriculum outlines the features of land investments.

© IFT. All rights reserved

	Raw land	Farmland	Timberland
Return drivers	Price of land	Harvest quantities Commodity prices Price of land	Biological growth Harvest quantities Lumber prices Price of land
Source of direct revenue	Price appreciation Lease revenue	Sale of crops and other agricultural products Price appreciation Lease revenue	Sale of trees, wood, and other timber products Price appreciation Lease revenue
Value	Physical location	Physical location Growth cycle Soil quality	Physical location Quality of timber Phase in timber production
Main risks	Best alternative use	Weather factors and climate change Biological factors, diseases	
Owners	Mostly institutional, some individual	Mostly individuals, some institutional	Mostly institutional, some individual
Ownership structure	Direct ownership, partnership	Direct ownership, partnership, REIT	Direct ownership, partnership, REIT, TIMO

Features and Forms of Farmland and Timberland Investment

Timberland tracts are typically thousands (or more) of acres in size, whereas farmland is frequently owned in smaller tracts of tens or hundreds of acres. As a result, farmland is more suited to family ownership, whereas timberland is more commonly owned by institutions.

An advantage of institutional ownership of physical farmland, as compared to exposure to crops through futures contracts is: Futures contracts are only available on a few common crops (such as wheat and corn). Ownership of physical farmland allows for the cultivation of crops that are not traded on futures exchanges, resulting in a broader universe of agricultural product price exposures.

However, the disadvantage is: there is limited price transparency or information to guide investment decisions; assistance from sector specialists is required. Direct investments are illiquid.

Farmland is highly sensitive to weather changes. Bad weather conditions can drastically reduce harvest yields.

© IFT. All rights reserved

3. Commodity Investment Forms

Commodities are physical products that can be standardized on quality, location, and delivery for investment purposes.

Generally, commodity investments take place through derivative instruments, because of the high storage and transportation costs incurred when holding commodities physically. The underlying asset of a commodity derivative may be a single commodity or an index of commodities. The return on commodity investment is based mainly on price changes rather than an income stream such as dividends.

In order to be transparent, investable, and replicable, commodity indexes typically use the price of the futures contracts rather than the prices of the underlying physical commodities.

Commodity sectors include:

- Energy - oil, natural gas, coal, electricity etc.
- Base metals - copper, aluminum, zinc etc.
- Precious metals - gold, silver, platinum etc.
- Agriculture - grains, livestock, coffee etc.
- Others - carbon credits, freight, forest products etc.

How are commodity futures contracts priced?

- The price of a futures contract can be calculated using the following formula:
  Future price $\approx$ Spot price $(1 + r)$ + Storage costs − Convenience yield
  where: convenience yield is the value associated with holding the physical asset;
  r is the short-term risk-free interest rate
- Future prices may be higher or lower than spot prices, based on convenience yield.
- For no arbitrage to occur, Future price ≈ Spot price (1+r). But commodities incur storage costs. So, they must be added to the future price and we get Future price ≈ Spot price (1 + r) + storage costs. Storage and interest costs are collectively known as "cost of carry".
- Why subtract the convenience yield? Because the buyer does not possess the commodity as of now, until the end of the contract. Since he has given up this convenience, it must be subtracted from the future price. That's how we arrive at Future price ≈ Spot price (1 + r) + storage costs - convenience yield
- Futures price may be higher or lower than the spot price based on the convenience yield.

  Contango: Future price > Spot price Markets tend to be in contango when there is little or no convenience yield.

  Backwardation: Future price < Spot price Markets tend to be in backwardation when the convenience yield is high.

© IFT. All rights reserved

Impact of inventory levels: When the inventory levels of a specific commodity are low, market participants would prefer to own the physical commodity rather than a derivative contract. This incentive raises spot prices relative to forward prices, resulting in a backwardation.

Forms of Commodity Investments

Commodity investments are typically made through derivatives as the storage and transportation costs for holding physical commodities are significant. Commodity derivative contracts may trade on exchanges or over the counter. The popular derivatives include futures, forwards, options, and swaps.

Commodity exposure can also be achieved through:

- Exchange traded products (either funds or notes): Suitable for investors seeking simplified trading through a standard brokerage account.
- Managed futures (also known as CTAs): Commodity trading advisers create trading strategies based on commodity derivative contracts that are aimed at predicting upcoming bull or bear trends.
- Funds that specialize in specific commodity sectors: e.g., private energy partnerships are similar to PE funds and can be used to gain exposure to the energy sector.

4. Natural Resource Investment Risk, Return, and Diversification

Commodities

Commodities offer potential for high returns, portfolio diversification, and inflation protection.

Commodity spot prices are a function of supply and demand, the costs of production and storage, value to users, and global economic conditions.

- Supplies of commodities depend on production and inventory levels.
- Demand of commodities depends on the consumption needs of end users.
- Demand may be high while supply may be low during economic growth; conversely, demand may be low and supply high during times of economic slowdown.
- If demand changes very quickly during any period, resulting in supply-demand mismatch, it may lead to price volatility.

Typically, commodity investments are made through derivative contracts, which are highly leveraged financial instruments. As a result, the observed returns are extremely volatile.

Commodities are attractive to investors not only for the potential profits but also because:

- Commodities exhibit high correlation with inflation over the last 30 years, suggesting that commodities are an effective inflation hedge. Some commodity prices are a component of inflation calculations e.g., food and energy.
- They provide effective portfolio diversification. Historically, the correlation between commodities and traditional investments has been low.

© IFT. All rights reserved

Farmland and Timberland

Timberland and farmland investments have similar risks as other real estate investments in raw land. However, weather is major risk factor for these investments. Droughts and flooding can dramatically decrease yields.

Another important risk factor is the international competitive landscape. Unlike other real estate that is mainly impacted by local factors, timberland and farmland produce commodities that are globally traded; therefore, they are impacted by global factors.

Farmland and timberland investments are traded infrequently and in private markets. As a result, despite the fact that both asset classes face significant risks, such as weather-related threats, they are likely to appear less volatile than commodities and other publicly traded risky assets (such as stocks).

ESG investors looking for responsible and sustainable investing can include timberland and farmland in their portfolios. These investments can help mitigate climate change.

Timberland and farmland have also exhibited low correlation with traditional investments. Thus, they can provide effective diversification benefits.

© IFT. All rights reserved

Summary

LO: Explain features of raw land, timberland, and farmland and their investment characteristics.

	Raw land	Farmland	Timberland
Return drivers	Price of land	Harvest quantities Commodity prices Price of land	Biological growth Harvest quantities Lumber prices Price of land
Source of direct revenue	Price appreciation Lease revenue	Sale of crops and other agricultural products Price appreciation Lease revenue	Sale of trees, wood, and other timber products Price appreciation Lease revenue
Value	Physical location	Physical location Growth cycle Soil quality	Physical location Quality of timber Phase in timber production
Main risks	Best alternative use	Weather factors and climate change Biological factors, diseases	
Owners	Mostly institutional, some individual	Mostly individuals, some institutional	Mostly institutional, some individual
Ownership structure	Direct ownership, partnership	Direct ownership, partnership, REIT	Direct ownership, partnership, REIT, TIMO

Timberland provides an income stream through the sale of trees, wood, and other timber products. Timberland can be thought of as both a factory and a warehouse. Additionally, since trees consume carbon as part of their life cycle, timberland considered a sustainable investment that mitigates climate-related risks.

Like timberland, farmland also provides an income component related to harvest quantities and agricultural commodity prices. However, it does not provide production flexibility, as farm products must be harvested when ripe.

LO: Describe features of commodities and their investment characteristics.

Generally, commodity investments take place through derivative instruments, because of the high storage and transportation costs incurred when holding commodities physically.

The return on commodity investment is based mainly on price changes rather than an income stream such as dividends.

© IFT. All rights reserved

LO: Analyze sources of risk, return, and diversification among natural resource investments.

Commodities are attractive to investors not only for the potential profits but also because:

- Commodities exhibit high correlation with inflation over the last 30 years, suggesting that commodities are an effective inflation hedge. Some commodity prices are a component of inflation calculations e.g., food and energy.
- They provide effective portfolio diversification. Historically, the correlation between commodities and traditional investments has been low.

Farmland and timberland investments are traded infrequently and in private markets. As a result, despite the fact that both asset classes face significant risks, such as weather-related threats, they are likely to appear less volatile than commodities and other publicly traded risky assets (such as stocks).

ESG investors looking for responsible and sustainable investing can include timberland and farmland in their portfolios. These investments can help mitigate climate change.

Timberland and farmland have also exhibited low correlation with traditional investments. Thus, they can provide effective diversification benefits.

© IFT. All rights reserved

LM06 Hedge Funds

1. Introduction....2
2. Hedge Fund Investment Features....2
3. Hedge Fund Investment Forms....5
4. Hedge Fund Investment Risk, Return, and Diversification....6
Summary....9

This document should be read in conjunction with the corresponding learning module in the 2024 Level I CFA® Program curriculum. Some of the graphs, charts, tables, examples, and figures are copyright 2023, CFA Institute. Reproduced and republished with permission from CFA Institute. All rights reserved.

Required disclaimer: CFA Institute does not endorse, promote, or warrant the accuracy or quality of the products or services offered by IFT. CFA Institute, CFA®, and Chartered Financial Analyst® are trademarks owned by CFA Institute.

Version 1.0

© IFT. All rights reserved

1. Introduction

This learning module covers:

- Investment features of hedge funds
- Investment forms and vehicles used in hedge fund investments
- Sources of risk, return and diversification benefits of hedge funds.

History: Hedge funds were originally started in 1949 as a way to hedge long-only stock portfolio. These funds followed three key principles:

- Always maintain short positions.
- Always use leverage.
- Only charge an incentive fee of 20% of the profits with no fixed fees.

Over time, the principles have changed. The following are the characteristics of hedge funds today:

- Aggressively managed portfolios of investments across asset classes and regions, use leverage, take long/short positions, and/or use derivatives.
- Generate high returns: either absolute or over a specified benchmark with minimal restrictions.
- Set up a private investment partnership with a limited number of investors who are willing to make a large initial investment.
- Investors are required to keep the money with the fund for a certain period – lockup period. Redemptions are not immediate. Usually, require a minimum notice period of 30 to 90 days.
- Invest anywhere there is a high return opportunity as restrictions are less.

2. Hedge Fund Investment Features

Hedge funds are private investment vehicles that pool money from institutions and high net worth individuals (accredited investors). Hedge funds typically have more flexible investment strategies than mutual funds and exchange-traded funds (ETFs).

Most hedge funds utilize some form of leverage (short selling, borrowing, derivatives, or sometimes all three) to enhance potential returns.

Hedge funds are not an asset class, but rather a collection of investment vehicles driven by a diverse set of investment strategies. Hedge funds are typically classified by strategy into five broad categories:

- **Event-driven:** A short term, bottom-up strategy that aims to profit from pricing inefficiencies before a major potential corporate event. Ex: bankruptcy, acquisition, merger, restructuring of a company, asset sale (large pocket of land in a prime location).

© IFT. All rights reserved

- **Relative value:** A strategy that seeks to profit from price discrepancy between related securities such as stocks and bonds.
- **Opportunistic:** Focuses on macro events and commodity trading.
- **Equity hedge:** Bottom-up strategy. Not focused on event-driven or macro strategies. Take long and short positions in publicly traded equity/equity derivative securities.
- **Multi-manager hedge funds:** Refers to a fund-of-fund hedge fund (explained later).

The sub-classifications under each category are listed below:

Sub-classification under event-driven category	
Merger Arbitrage	• Go long (buying) on the stock of the company being acquired and go short on the stock of the acquiring company. • Risk: many corporate events such as merger do not occur as planned and if the fund has not closed its positions on time, it may incur losses.
Distressed/restructuring	• Purchase and profit from debt securities of companies that are either in bankruptcy or near bankruptcy. • Strategy: the fixed income securities would be priced at a significant discount to their par value; these can be sold later at a profit at settlement (liquidation or equity stake) • Other complicated strategies: Buy senior debt/short junior debt. • Buy preferred stock/short common stock.
Activist	• Purchase a managing equity stake in a public company that is believed to be mismanaged, and then influence its policies. • May advocate restructure, changes in strategy, hiving off non-profitable units, etc.
Special Situations	• Purchase equity of companies engaged in restructuring activities other than merger/bankruptcy.
Sub-classification under relative value category	
Fixed-Income Convertible Arbitrage	• A market neutral strategy to exploit mispricing in convertible bond and issuer's stock. • Long position in convertible debt + short position in issuer's common stock. • As the name implies, it has a theoretical zero-beta portfolio. *Note: A convertible bond is a bond (hybrid security) that can be converted into common stock at a pre-determined price at a*

© IFT. All rights reserved

	pre-determined time. Usually, the yield is lower than a comparable bond.
Fixed-Income Asset Backed	• Exploit mispricing of asset-backed securities.
Fixed-Income General	• Exploit mispricing between two corporate issuers (i.e. long/short trades), between corporate and government issuers, or between different parts of the same issuer's capital structure.
Volatility	• Go long or short market volatility within a specific asset class.
Multi-Strategy	• Generate consistently absolute positive returns irrespective of how the equity, debt, or currency markets are performing. • Does not focus on one strategy, but allocates capital across different strategies where investment opportunities exist. Ex: equity long/short, convertible arbitrage, merger arbitrage, etc. • Unlike funds of funds, multi-strategy funds execute strategies within one fund group and they do not have the extra layer of fees associated with a fund of funds.
Sub-classifications under opportunistic strategies	
Macro strategies	• Uses a top-down approach to identify trends based on changes in economic policies across the globe. • The strategies could focus on currency markets, fixed income markets, or based on changes in interest rates. • Trades are based on expected movement in economic variables.
Managed futures	• Actively managed funds that make diversified directional investments in futures markets using technical and fundamental strategies. • Because they have historically focused on commodity futures, managed futures funds are also known as commodity trading advisers (CTAs).
Sub-classifications under the equity hedge category	
Fundamental long/short	• The hedge fund takes long positions in undervalued securities and short positions in overvalued securities. • The manager typically maintains a net long exposure but may adjust the amount of net market risk depending on his or her market forecast.

© IFT. All rights reserved

Market Neutral	• Uses quantitative/fundamental analysis to identify undervalued/overvalued securities. • **Strategy**: buy (long) undervalued securities and sell (short) overvalued securities. Hold equal dollar amounts in both positions. • Neutral with respect to market risk, i.e., the portfolio beta is close to zero.
Fundamental Growth	• Uses fundamental analysis to identify companies with high growth potential and capital appreciation. • **Strategy**: long position in such stocks.
Fundamental Value	• Uses fundamental analysis to identify undervalued companies. • **Strategy**: long position in such stocks.
Short Bias	• Uses quantitative/fundamental analysis to identify overvalued securities. • **Strategy:** short position in overvalued securities.

Distinguishing Characteristics of Hedge Fund Investments

The key characteristics of hedge funds that distinguish them from traditional investments are:

1. Fewer legal and regulatory constraints.
2. Flexible mandates that allow the use of shorting and derivatives.
3. A larger investment universe to choose from.
4. Aggressive investment styles that permit concentrated positions in securities offering exposure to credit, volatility, and liquidity risk premiums.
5. Relatively liberal use of leverage.
6. Liquidity constraints such as lockups and liquidity gates.
7. Relatively high fee structures with management and incentive fees.

3. Hedge Fund Investment Forms

Direct Hedge Fund Investment Forms

The majority of hedge funds are structured as limited partnerships, with the portfolio manager serving as the general partner (GP) and the institutional investors serving as limited partners (LPs). This is referred to as the direct form of hedge fund setup.

A common hedge fund fee structure is “2 and 20” – the hedge fund manager receives a 2% management fee on the fund’s AUM and a 20% if the fund’s net profit. Recently, due to pressure from investors, hedge funds now offer a “1 or 30” fee structure – where the manger

© IFT. All rights reserved

receives the greater of a 1% management fee or an incentive fee of 30% of the fund's outperformance against a benchmark (instead of a performance fee based simply on total profits.)

The fund offering documents govern the legal and contractual relationship between GPs and LPs. Additionally, a manager could create a "side letter" that is only applicable to certain investors and has different legal, regulatory, tax, operational, or reporting requirements.

For larger investors, the hedge fund could be structured as a fund of one or a separately managed account (SMA). In the case of a fund of one structure, the hedge fund is created for one investor. In the case of an SMA, the investor creates his or her own investment vehicle and the underlying assets are held and registered in investor's name. However, the day-to-day management of the account is delegated to the hedge fund manager.

An SMA structure allows for a customizable portfolio with investor-specific investment mandates, improved transparency, efficient capital allocation, and increased liquidity, allowing the investor to exercise greater control while keeping the fees low.

Indirect Hedge Fund Investment Forms

Selecting and investing in an individual hedge fund can be difficult because of the extensive due diligence required and the high minimum investment threshold. Therefore, some investors may prefer to invest in a fund of hedge funds.

A diversified portfolio of hedge funds is often referred to as a **fund of funds**. This instrument makes hedge funds accessible to smaller investors or to those who do not have the resources, time, or expertise to choose among hedge fund managers. Other benefits include:

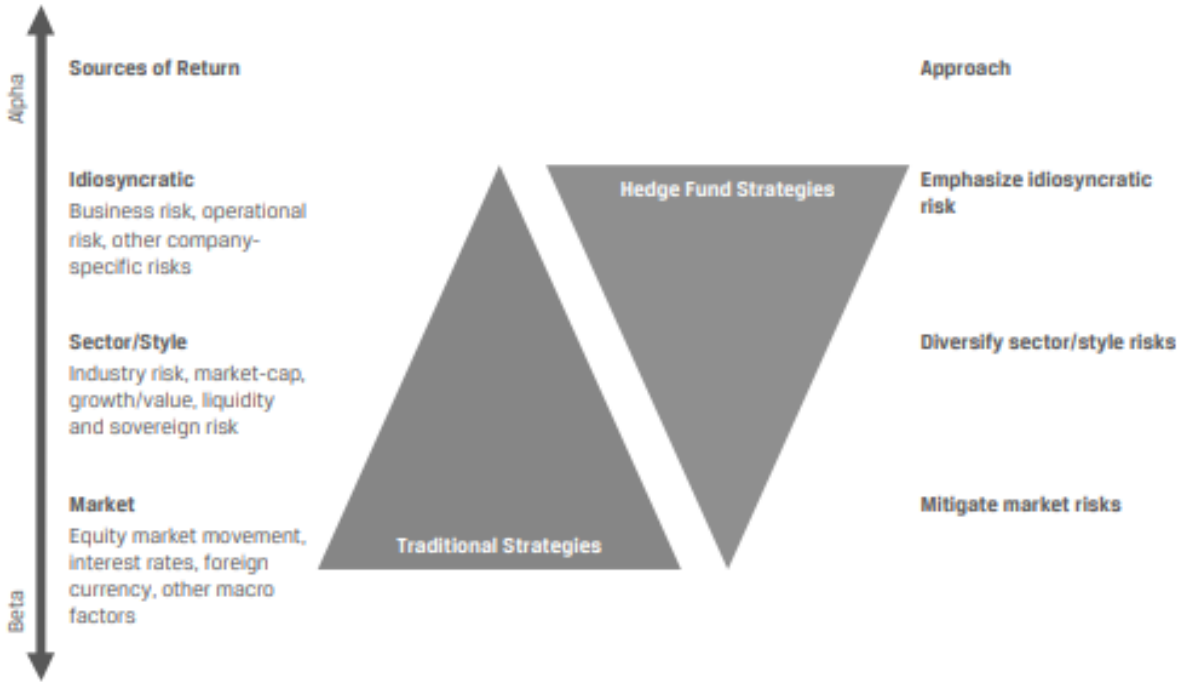
- Better redemption terms
- Due diligence expertise
- More diversification as they invest in hedge funds across geographies and strategies

However, fund of funds may charge an additional 1% management fee and 10% incentive fee on top of the fees charged by the underlying hedge funds. This double layer of fees can significantly reduce the after fee returns to the investor.

4. Hedge Fund Investment Risk, Return, and Diversification

As shown in Exhibit 5 of the curriculum, hedge funds have a different approach towards return generation as compared to traditional investments. Hedge funds seek to limit market exposure and returns from beta and primarily focus on generating idiosyncratic returns by identifying sources of unique return, or alpha. The primary source of hedge fund excess return is market inefficiencies and the manager's skill to capitalize on them.

© IFT. All rights reserved



Several hedge fund indexes are created sing publicly available hedge fund data. However, these indexes suffer from the following biases:

- Self-reporting bias: Only high-performing funds are likely to make data available to the public.
- Survivorship bias: Hedge funds that have stopped reporting are removed from the index.
- Backfill bias: When a new hedge fund starts reporting for the first time, its past performance is also added to the index. It is very likely that the fund began reporting because it had a stellar past performance.

These biases cause the index return to be overstated as compared to the actual performance.

Historically, hedge funds have provided higher returns than either stocks or bonds with a relatively low standard deviation. They have certainly added value to institutional investors as a portfolio diversifier.

Diversification Benefits of Hedge Fund Investments

Hedge funds use several strategies such as market-neutral, relative value, and event-driven strategies to achieve diversification benefits and outperform equity markets on a risk-adjusted basis.

It is believed that less-than-perfect correlation of hedge funds with stocks provides diversification benefit. However, during financial crisis periods, the correlation between hedge fund performance and stock market performance may increase.

© IFT. All rights reserved

Between 2009 – 2019, most hedge funds failed to beat the performance of equity and bonds. But they still continue to be a part of institutional asset allocations because of their risk-diversification properties.

© IFT. All rights reserved

Summary

LO: Explain investment features of hedge funds and contrast them with other asset classes.

Hedge funds are private investment vehicles that pool money from institutions and high net worth individuals (accredited investors).

Hedge funds are typically classified by strategy into five broad categories (as shown in Exhibit 1 from the curriculum):

Equity	Event Driven	Relative Value	Opportunistic	Multi-Manager
Long/Short Equity	Merger Arbitrage	Convertible Bond Arbitrage	Global Macro	Fund of Funds
Short Biased	Distressed	Fixed-Income Arbitrage	Managed Futures	
Market Neutral	Special Situations	Multi-Strategy		
	Activist			

The key characteristics of hedge funds that distinguish them from traditional investments are:

1. Fewer legal and regulatory constraints.
2. Flexible mandates that allow the use of shorting and derivatives.
3. A larger investment universe to choose from.
4. Aggressive investment styles that permit concentrated positions in securities offering exposure to credit, volatility, and liquidity risk premiums.
5. Relatively liberal use of leverage.
6. Liquidity constraints such as lockups and liquidity gates.
7. Relatively high fee structures with management and incentive fees.

LO: Describe investment forms and vehicles used in hedge fund investments.

The majority of hedge funds are structured as limited partnerships, with the portfolio manager serving as the general partner (GP) and the institutional investors serving as limited partners (LPs). This is referred to as the direct form of hedge fund setup.

For larger investors, the hedge fund could be structured as a fund of one or a separately managed account (SMA).

For smaller and retail investors, indirect forms, such as funds of funds, help obtain a hedge fund exposure.

© IFT. All rights reserved

LO: Analyze sources of risk, return, and diversification among hedge fund investments.

Hedge funds seek to limit market exposure and returns from beta and primarily focus on generating idiosyncratic returns by identifying sources of unique return, or alpha. The primary source of hedge fund excess return is market inefficiencies and the manager's skill to capitalize on them.

Hedge funds use several strategies such as market-neutral, relative value, and event-driven strategies to achieve diversification benefits and outperform equity markets on a risk-adjusted basis.

© IFT. All rights reserved

LM07 Introduction to Digital Assets

1. Introduction....2
2. Distributed Ledger Technology....2
3. Digital Asset Investment Features....6
4. Digital Asset Investment Forms....7
5. Digital Asset Investment Risk, Return, And Diversification....10
Summary....12

This document should be read in conjunction with the corresponding learning module in the 2024 Level I CFA® Program curriculum. Some of the graphs, charts, tables, examples, and figures are copyright 2023, CFA Institute. Reproduced and republished with permission from CFA Institute. All rights reserved.

Required disclaimer: CFA Institute does not endorse, promote, or warrant the accuracy or quality of the products or services offered by IFT. CFA Institute, CFA®, and Chartered Financial Analyst® are trademarks owned by CFA Institute.

Version 1.0

1. Introduction

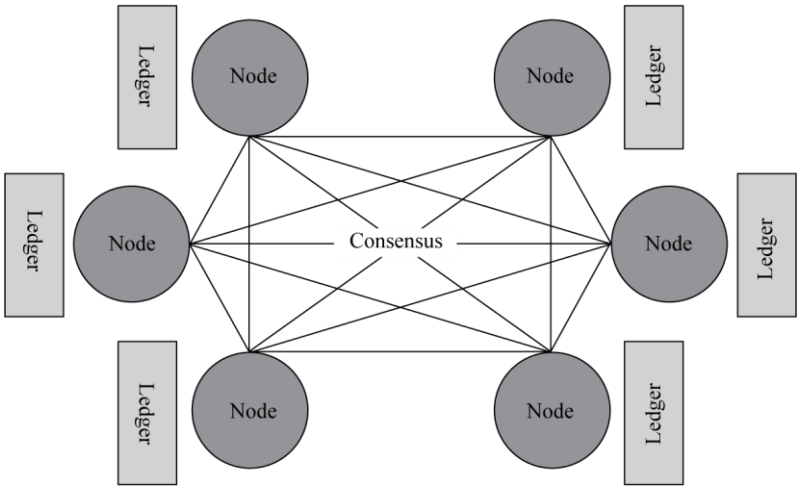
Digital assets are a relatively new investment class consisting of assets that can be created, stored, and transmitted electronically and have ownership or use rights associated with them.

This learning module covers:

- Distributed ledger technology
- Investment features of digital assets
- Investment forms and vehicles used in digital asset investments
- Investment risk, return, and diversification benefits of digital asset investments

2. Distributed Ledger Technology

A **distributed ledger** is a database which can be shared across computer entities (or nodes) in a network. This is illustrated in the following diagram.



There can be a potentially infinite number of nodes in a network. Every node will have a copy of the distributed ledger. There is a consensus mechanism which ensures that all these ledgers are kept in sync. Through the consensus mechanism all nodes agree on a new transaction and update their ledgers. New records are considered immutable, which means once a record is created it cannot be changed.

Distributed ledger technology (DLT) uses **cryptography**, which refers to encrypting and decrypting data. Through encryption, we ensure that the data remains secure.

DLT also accommodates **smart contracts**. These are computer programs that self-execute on the basis of pre-specified terms and conditions. For example, contracts that automatically transfer collateral from the borrower to the lender in the event of default.

© IFT. All rights reserved

DLT networks allow us to create, exchange, and track ownership of financial assets on a peer-to-peer basis. There is no central authority to validate the transactions.

DLT benefits include:

- Accuracy, transparency, and security in the record keeping process.
- Faster transfer of ownership.
- Peer-to-peer interactions.

A drawback of DLT is that the computational processes underlying DLT require massive amounts of energy to verify transaction activity.

Blockchain is a type of distributed ledger. Its characteristics are:

- Information is recorded sequentially within blocks.
- Blocks are chained and secured using cryptography.
- Each block contains a grouping of transactions and a secure link to the previous block.

The following steps outline the process of adding new transactions to the Blockchain network.

1. A transaction takes place between buyer and seller.
2. The transaction is broadcast to the network of computers (nodes).
3. The nodes validate the transaction details and parties to the transaction.
4. Once verified, the transaction is combined with other transactions to form a new block (of predetermined size) of data for the ledger.
5. This block of data is then added or linked (using a cryptographic process) to the previous block(s) containing data.
6. The transaction is considered complete and the ledger has been updated.

Proof of Work vs. Proof of Stake

A consensus protocol is a set of rules that govern how blocks in a blockchain network are cryptographically chained together for the verification of the complete and immutable history of transaction records.

Consensus protocols are classified into two types: "proof of work" (PoW) and "proof of stake" (PoS).

The Proof of Work Protocol

The proof of work protocol uses a computationally expensive lottery to determine which specific block to add. To verify a transaction, the PoW consensus mechanism employs a cryptographic problem that must be solved by some computers on the network (known as miners) each time a transaction occurs.

Miners use powerful computers and large amounts of energy to solve complex algorithm puzzles in order to validate and lock blocks of transactions into the blockchain and earn cryptocurrency in the process. The "proof of work" consensus process for updating the

© IFT. All rights reserved

blockchain can require significant amounts of computing power, making it extremely difficult and expensive for a single third party to manipulate historical data.

The Proof of Stake Protocol

This protocol requires selected network participants, known as validators, to pledge capital in order to vouch for the validity of a block. This stake signals the network that a validator is available to verify the veracity of a transaction and propose a block.

A majority of the other validators who have staked a digital asset in the network must then attest to the proposed block's validity. Validators benefit from both proposing and attesting to the validity of blocks proposed by other participants.

The security standards of the PoS protocol are based on a group of stakeholders (and their pledged stake) controlling the network's computational power and protecting access from malicious parties gaining a majority.

Permissioned and Permissionless Networks

DLT can take the form of permissionless or permissioned networks.

Permissionless networks are open to any user who wishes to make a transaction. Once a transaction is added, it cannot be changed. All users can see all transactions on the block chain. These networks do not depend on a central authority. Bitcoin is a popular example of an open permissionless network.

In **permissioned** networks, network members may be restricted from participating in certain network activities. Controls or permissions might be used. Different users may have different levels of access to the ledger. For example, participants may be allowed to enter transactions while regulators may be allowed to view the transaction history.

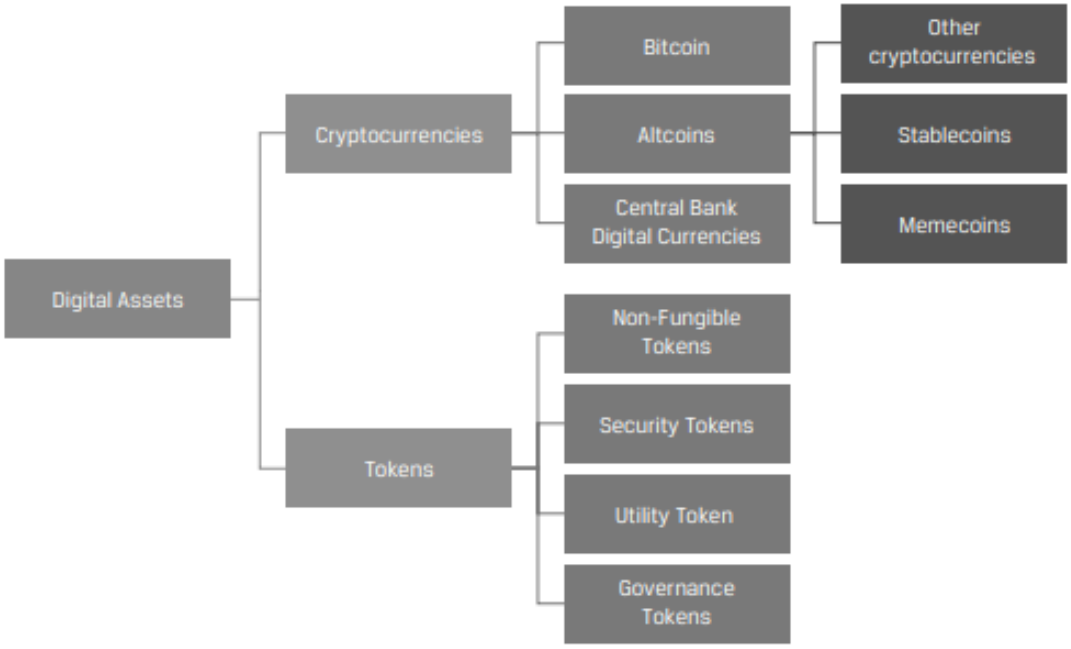
Exhibit 3 from the curriculum compares the features of permissioned and permissionless blockchains.

	Permissioned blockchain	Permissionless blockchain
Speed	Faster as only a limited number of members participate or are authorized to validate transactions	Slower as a large number of members have to reach consensus, which decreases network speed and scalability
Cost	Cost-effective as few members are required to validate each transaction	Not cost-effective as many members are required to validate each transaction
Decentralization	Partially decentralized as there are a limited number of members in the chain	Decentralized as all members can access the network
Access	Membership is limited.	Membership is unlimited.
Governance	The governance is determined by a centralized organization.	The governance is decentralized and is maintained by the members.

© IFT. All rights reserved

Types of Digital Assets

Digital assets are assets that exist only as an electronic record with the right to use, buy, or sell them. Exhibit 4 from the curriculum shows the different types of digital assets.



Cryptocurrencies:

They are also called digital currency or electronic currency. They do not have any physical form, but allow transactions to take place between buyers and sellers. They are issued by private individuals or organizations. There is no central authority, like a central bank backing these currencies.

Many cryptocurrencies have a self-imposed limit on the total amount of currency they may issue. For example, a well-known cryptocurrency, Bitcoin has a self-imposed limit of 21 million. Such limits could help maintain a value from a technical perspective, yet we should also recognize the fact that with cryptocurrencies there is a lack of fundamentals underlying the value of the currency. Hence, they tend to be very volatile relative to major currencies like the Dollar or the Euro.

An *initial coin offering* (ICO) is an unregulated process whereby companies sell their crypto-tokens to investors. Through this process, investors fund the company and the tokens can be used to buy products and services from the company at a latter point in time.

Central banks across the world are recognizing the potential benefits and examining use cases for their own cryptocurrency versions as a substitute to physical currency. Central bank digital currencies (CBDCs) are typically designed as a tokenized version of the central bank's currency ("fiat currency")—essentially, a digital bank note or coin.

© IFT. All rights reserved

Tokenization:
It is the process of representing ownership rights to physical assets on a blockchain or distributed ledger. Usually transactions involving physical assets, such as real estate, require substantial efforts in ownership verification and examination. DLT can streamline this process by creating a single digital record of ownership.

Another type of digital asset is a **non-fungible token (NFT).** Using blockchain technology, an NFT connects digital assets to certificates of authenticity. NFTs differ from "fungible" tokens like cryptocurrencies in that each token and the authenticated object it represents are unique. As a result, they can "stamp" assets and represent digital assets in a virtual world. The most common application for NFTs is the trading of digital artwork.

Security tokens digitize the ownership rights associated with publicly traded securities. The custody of these security tokens can be stored on a blockchain, which can increase the efficiency of post trade processing and settlement. The current post-trade clearing and settlement in the financial securities market is quite cumbersome. Security tokens have the ability to streamline this process by providing near-real-time trade verification, reconciliation, and settlement. This can significantly reduce the complexity, time, and cost involved with processing transactions.

Utility tokens provide services within a network, such as pay for services and network fees. While security tokens may pay dividends, utility tokens only compensate for activities on the network.

Governance tokens are important in permissionless networks. They act as a vote to determine how specific networks are run.

3. Digital Asset Investment Features

Digital assets are frequently regarded as an alternative asset class. As digital assets mature, institutional investors may continue to seek exposure to these assets for higher returns and potential diversification benefits.

Distinguishing Characteristics of Digital Assets

Digital assets differ from traditional financial assets in terms of their inherent value, transaction validation approach, uses as a legal medium of exchange, and legal and regulatory protection. Exhibit 5 from the curriculum summarizes these differences.

© IFT. All rights reserved

	Digital Assets	Traditional Financial Assets
Inherent value	▪ No fundamental value or future cash flow generation ▪ Price driven by certain features on the blockchain	▪ Value determined by future cash flow generated from the assets
Transaction validation	▪ Usually recorded on decentralized digital ledgers using cryptography and algorithms for permissionless blockchain networks	▪ Recorded in private ledgers maintained by central intermediaries
Uses as a medium of exchange	▪ Very few digital assets are used as a direct medium of exchange, mainly targeting large-scale commercially viable acceptance	▪ Not used directly as a medium of exchange but can be readily transacted and exchanged into traditional fiat currencies that are widely used in the real world
Legal and regulatory protection	▪ Ambiguous, often contradictory, evolving framework; generally unregulated, with minimal legal protections ▪ Use can be illegal or criminal in some countries	▪ Well-established, tested, and proven legal, regulatory, and commercial standards that are clear, predictable, and well defined across all jurisdictions

Investible Digital Assets

The best-known and widely traded cryptocurrency is 'Bitcoin'. It was designed as an alternative to traditional currencies: as a medium of exchange and store of value.

Several other cryptocurrencies exist on technology similar to Bitcoin. These are called altcoins.

Altcoins

The most popular altcoin is 'Ether'. It was launched in 2015 on its own Ethereum network. It includes an additional feature of a programmable blockchain, which enables users to build applications that use the blockchain to validate or secure transactions. As a result, ether and other programmable coins have the potential to be used for more than just storing value. Such programmable altcoins are also known as smart coins.

Stablecoins

Stablecoins are designed to maintain a stable value by linking their value to another asset and are collateralized by a basket of assets such as fiat currencies, precious metals, or other cryptocurrency. The reserve basket protects holders from price volatility and reduces the risk to stablecoin holders if the cryptocurrency experiences transaction problems.

Meme Coins

Meme coins are cryptocurrencies that are often inspired by a joke and are generally launched for entertainment purposes because they gain popularity quickly, allowing early purchasers to sell their holdings at a significant profit. A popular meme coin is Dogecoin.

4. Digital Asset Investment Forms

Investment in digital assets can be in the form of direct investment on the blockchain or indirect investments through exchange-traded products and hedge funds.

© IFT. All rights reserved

Direct ownership of Bitcoin and other cryptocurrencies is accomplished through the use of a **cryptocurrency wallet**, which stores the (public and private) digital codes required to access the asset via a computer website or mobile device application.

Cryptocurrency exchanges can be classified into centralized exchanges and decentralized exchanges.

Centralized exchanges:

- Although incompatible with Bitcoin's decentralized ideology, this is the most popular type of exchange.
- Privately held exchanges provide trading platforms for cryptocurrencies and offer volume, liquidity, and price transparency.
- Trading is electronic and direct, with no intermediary broker or dealer, and takes place on private servers, exposing the centralized exchanges and their clients to security risks.
- If the exchange's servers are compromised, the entire system may be rendered inoperable, halting trade, and leaking sensitive user information.
- Some exchanges may be regulated depending on the jurisdiction.

Decentralized exchanges:

- These exchanges mimic blockchain's decentralized protocol and operate similarly to how Bitcoin operates.
- If one of the computers on the network is attacked, the exchange remains operational because there are numerous other computers that continue to operate.
- Therefore, attacking decentralized exchanges is substantially more difficult.
- However, decentralized exchanges are difficult to regulate because no single individual, organization, or group controls the system.

Both centralized and decentralized exchanges are vulnerable to fraud and manipulation, because they are not subject to rigorous oversight and are generally not regulated as financial exchanges.

Direct Digital Asset Investment Forms

Direct investment in cryptocurrencies has several risks:

- The risk of fraud such as scam IPOs, various pump and dump schemes, market manipulation, theft, and schemes that seek to hack access credentials.
- Cryptocurrency wallets can only be accessed with a unique passkey. When the passkey is lost, the wallet's contents are lost forever. Approximately 20% of all Bitcoins are said to be in lost or abandoned wallets that their owners cannot access.

© IFT. All rights reserved

- Many of the smaller cryptocurrencies may be held primarily by a small group of holders known as "whales." The cryptocurrency community uses the term "whales" to refer to individuals or entities who own a large enough amount of a cryptocurrency to be able to manipulate its price.

Indirect Digital Asset Investment Forms

Alternatives to gain indirect exposure to digital assets include:

Cryptocurrency coin trusts:

- These trusts hold large pools of a cryptocurrency and allow investors to trade in shares of the trust (like a closed-end fund).
- Pros: The investor does not need to create a separate digital wallet. Also, the trust may provide transparency into trading.
- Cons: Trusts charge substantial fees, sometimes more than 2%. The trust shares may trade at a premium or discount to their NAV.

Cryptocurrency futures:

- These are agreements to buy or sell a specific quantity of a cryptocurrency at a specified price on a particular date. Ex: Bitcoin futures on Chicago Mercantile Exchange (CME).
- These contracts are typically cash settled with no actual cryptocurrency changing hands.
- Cons: Futures contracts are inherently leveraged. The cryptocurrency futures market may be less developed, less liquid, and more volatile than more established futures markets.

Cryptocurrency exchange-traded funds:

- Many ETFs that seek to replicate the returns of digital assets have recently been launched.
- These ETFs typically do not invest directly in cryptocurrencies and instead gain exposure to the value of cryptocurrencies through the use of cash and cryptocurrency derivatives.

Cryptocurrency stocks: Examples include equity in publicly traded –

- digital exchanges.
- payment providers accepting cryptocurrencies.
- corporations accepting cryptocurrencies as payments, investing in cryptocurrencies, or mining cryptocurrencies.

© IFT. All rights reserved

- corporations developing and/or manufacturing products or services that are used for running blockchain networks, such as specialized computers used for mining.

Hedge funds investing in cryptocurrencies:

- Several discretionary long, long/short, quantitative, and multi-strategy funds provide exposure to cryptocurrencies.
- Some hedge funds actively mine for Bitcoin to generate further returns.

Digital Forms of Investment for Non-Digital Assets

Asset-backed tokens are digital claims on physical assets, financial assets, or financial instruments and are collateralized by these underlying assets. Examples of tokenized assets include: gold, crude oil, real estate, and equities.

Asset-backed tokens have the potential to increase liquidity by allowing for fractional ownership of high-priced assets such as houses, art, precious metals, and precious stones, allowing multiple investors to own a fractional interest in the same asset. The digital representation of ownership allows for an immutable record of ownership information and ownership transfer, increasing transparency and lowering transaction, intermediation, and record-keeping costs.

Asset-backed tokens are often issued on the Ethereum network or other smart contract platforms that enable peer-to-peer interaction via interoperable, transparent smart contracts that last for the duration of the chain. These decentralized applications, or dApps, enable transactions to occur and be recorded on the blockchain without the need for a central coordinating mechanism.

The push for financial decentralized applications based on opensource codes and smart contracts has grown into a movement known as decentralized finance, or DeFi. DeFi seeks to design, combine, and develop decentralized financial applications as building blocks for sophisticated financial products and services.

5. Digital Asset Investment Risk, Return, and Diversification

Digital Asset Investment Risks and Returns

Bitcoin and other digital assets are priced based on future asset appreciation rather than any underlying cash flow. The market demand for the limited supply of cryptocurrencies is a significant driver of prices. For example, the supply of Bitcoin is limited to 21 million Bitcoins, by design. Due to this some investors view Bitcoins as the digital version of gold.

Bitcoin's performance has been characterized by high return, high volatility and low correlations with traditional asset classes. Bitcoin's price rose from $0.05 at its inception to a historical high of $68,789 on 10 November 2021, before crashing to around $17,709 on 18 June 2022. Early Bitcoin investors were rewarded handsomely with phenomenal price appreciation.

© IFT. All rights reserved

However, as the brief track record shows cryptocurrencies can be very volatile and risky investments. Several countries have imposed severe restrictions on trading and ownership of cryptocurrencies, ex: China, which banned the asset in 2021.

Diversification Benefits of Digital Asset Investments

Because of their historically low correlations with other asset classes, digital assets can help diversify a traditional asset portfolio. However, correlations have been observed to increase, particularly during periods of high market uncertainty.

© IFT. All rights reserved

Summary

LO: Describe financial applications of distributed ledger technology.

A distributed ledger is a database which can be shared across computer entities (or nodes) in a network.

DLT networks allow us to create, exchange, and track ownership of digital assets on a peer-to-peer basis. There is no central authority to validate the transactions.

DLT could also bring efficiencies to post-trade and compliance processes through automation, smart contracts, and identity verification.

DLT can take the form of permissionless or permissioned networks.

A consensus protocol is a set of rules that govern how blocks in a blockchain network are cryptographically chained together for the verification of the complete and immutable history of transaction records. Consensus protocols are classified into two types: "proof of work" (PoW) and "proof of stake" (PoS).

LO: Explain investment features of digital assets and contrast them with other asset classes.

The different types of digital assets are shown in the figure below:

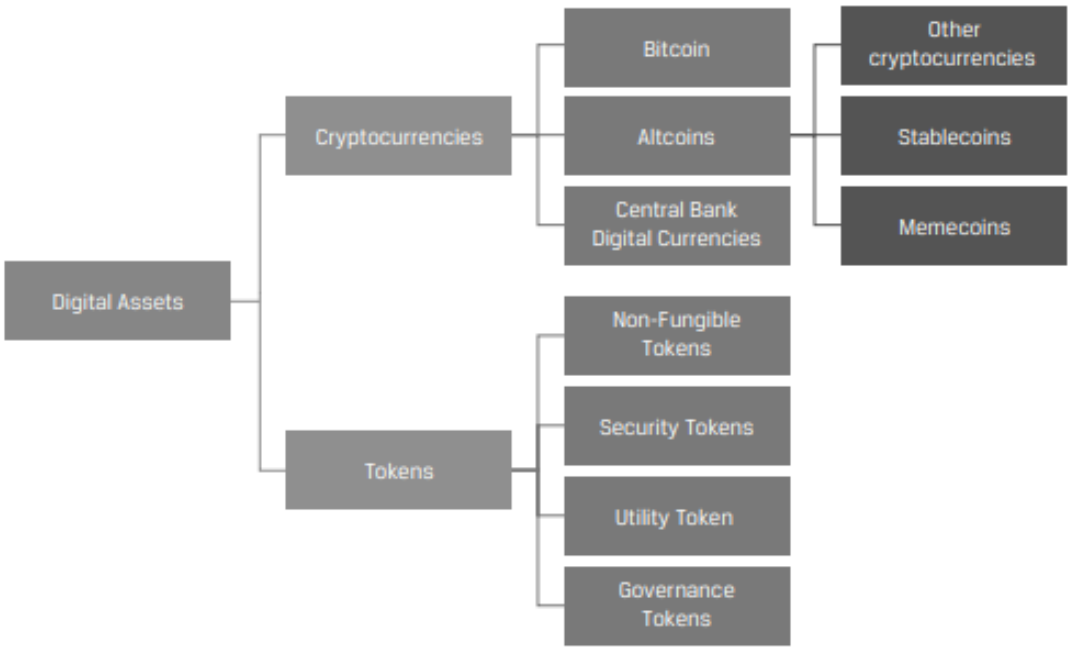


Exhibit 5 from the curriculum summarizes the differences between digital assets and traditional financial assets.

© IFT. All rights reserved

	Digital Assets	Traditional Financial Assets
Inherent value	▪ No fundamental value or future cash flow generation ▪ Price driven by certain features on the blockchain	▪ Value determined by future cash flow generated from the assets
Transaction validation	▪ Usually recorded on decentralized digital ledgers using cryptography and algorithms for permissionless blockchain networks	▪ Recorded in private ledgers maintained by central intermediaries
Uses as a medium of exchange	▪ Very few digital assets are used as a direct medium of exchange, mainly targeting large-scale commercially viable acceptance	▪ Not used directly as a medium of exchange but can be readily transacted and exchanged into traditional fiat currencies that are widely used in the real world
Legal and regulatory protection	▪ Ambiguous, often contradictory, evolving framework; generally unregulated, with minimal legal protections ▪ Use can be illegal or criminal in some countries	▪ Well-established, tested, and proven legal, regulatory, and commercial standards that are clear, predictable, and well defined across all jurisdictions

LO: Describe investment forms and vehicles used in digital asset investments.

Investment in digital assets can be in the form of direct investment on the blockchain or indirect investments through exchange-traded products and hedge funds.

Direct ownership of Bitcoin and other cryptocurrencies is accomplished through the use of a cryptocurrency wallet. Cryptocurrency exchanges can be classified into centralized exchanges and decentralized exchanges.

Alternatives to gain indirect exposure to digital assets include:

- Cryptocurrency coin trusts
- Cryptocurrency futures
- Cryptocurrency exchange traded funds
- Cryptocurrency stocks
- Hedge funds investing in cryptocurrency

Asset-backed tokens are digital claims on physical assets, financial assets, or financial instruments and are collateralized by these underlying assets.

The push for financial decentralized applications based on opensource codes and smart contracts has grown into a movement known as decentralized finance, or DeFi. DeFi seeks to design, combine, and develop decentralized financial applications as building blocks for sophisticated financial products and services

LO: Analyze sources of risk, return, and diversification among digital asset investments.

Bitcoin and other digital assets are priced based on future asset appreciation rather than any underlying cash flow. The market demand for the limited supply of cryptocurrencies is a

© IFT. All rights reserved

significant driver of prices.

Bitcoin's performance has been characterized by high return, high volatility and low correlations with traditional asset classes.

Because of their historically low correlations with other asset classes, digital assets can help diversify a traditional asset portfolio. However, correlations have been observed to increase, particularly during periods of high market uncertainty.

© IFT. All rights reserved